

Inteligencia de Negocios

Crear un proyecto y un paquete básico con SSIS

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y Valderrama



Índice

Temas a presentar

Objetivos de la lección

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ETL

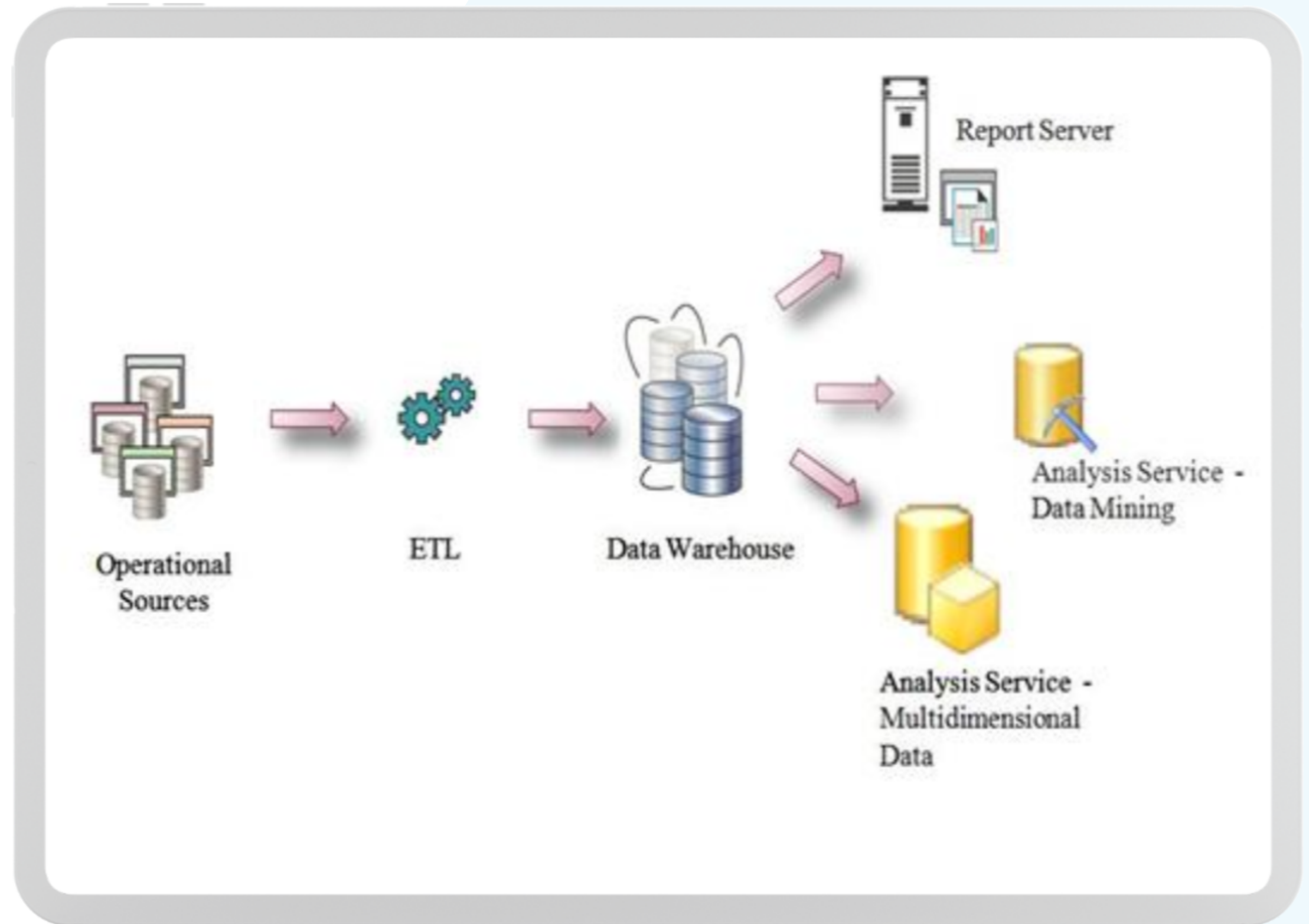
Extract, transform and load



Microsoft®
SQL Server®
Integration Services

El Objetivo

Reformatear la data a
una tabla propia...



Relación de la lección con el curso de BI



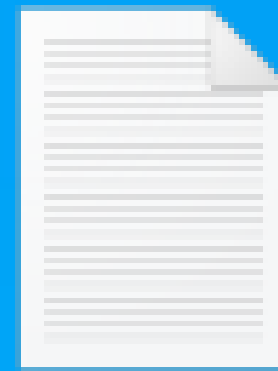
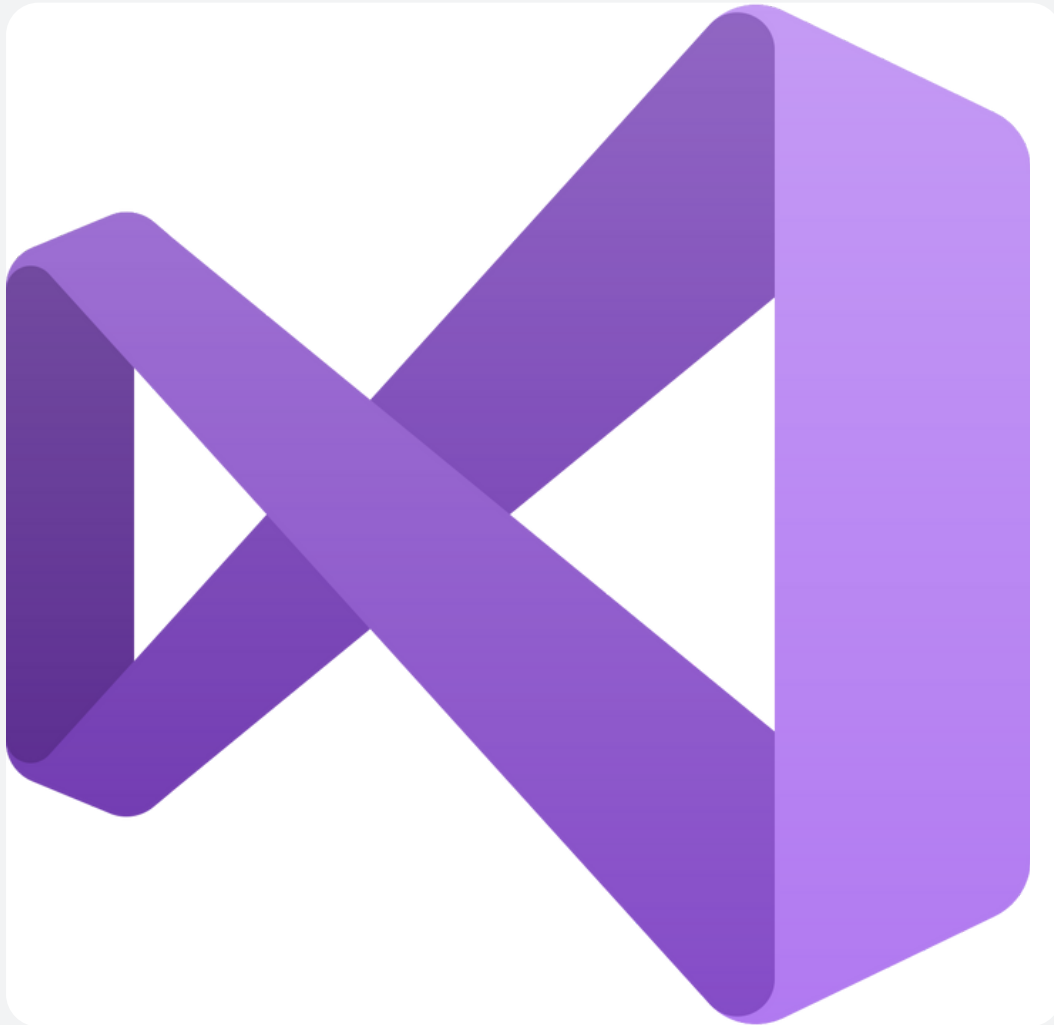
CARACTERÍSTICAS DE SQL SERVER BUSINESS INTELLIGENCE

El poder clave de SSIS es su capacidad de transformación y migración de datos. SSIS proporciona herramientas de almacenamiento de datos como la extracción y la transformación de datos.

ETLs

ETL (Extract, Transform, Load) es un proceso que consiste en extraer datos de diferentes fuentes de datos, manipularlos según los cálculos del negocio y cargar los datos modificados en un almacén de datos diferente.

Prerequisites



SampleCurrencyData.txt

Data Warehouse

[AdventureWorksDW2019.bak](#) ↗

[AdventureWorksDW2017.bak](#) ↗

[AdventureWorksDW2016.bak](#) ↗

[AdventureWorksDW2016_EXT.bak](#) ↗

[AdventureWorksDW2014.bak](#) ↗

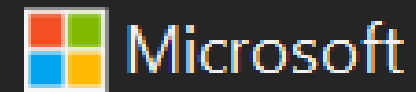
[AdventureWorksDW2012.bak](#) ↗

Prerequisite 1



SQL Server Integration Services Projects

This project may be used for building high performance data integration and workflow solutions, including extraction, transfor...



Modifying — Visual Studio Community 2019 — 16.11.19

Workloads

Individual components

Language packs

Installation locations

Windows using C++.

Gaming (2)



Game development with Unity

Create 2D and 3D games with Unity, a powerful cross-platform development environment.



Game development with C++

Use the full power of C++ to build professional games powered by DirectX, Unreal, or Cocos2d.



Other Toolsets (6)



Data storage and processing

Connect, develop, and test data solutions with SQL Server, Azure Data Lake, or Hadoop.



Data science and analytical applications

Languages and tooling for creating data science applications, including Python and F#.



Installation details

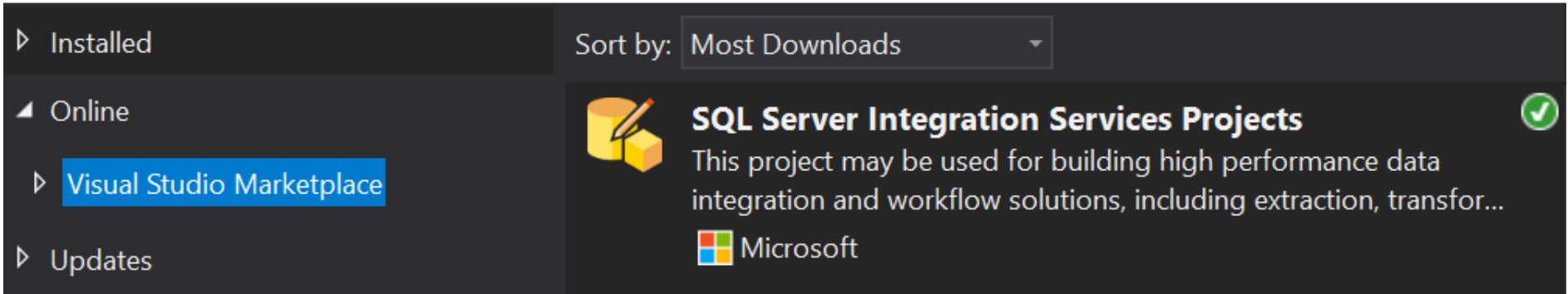
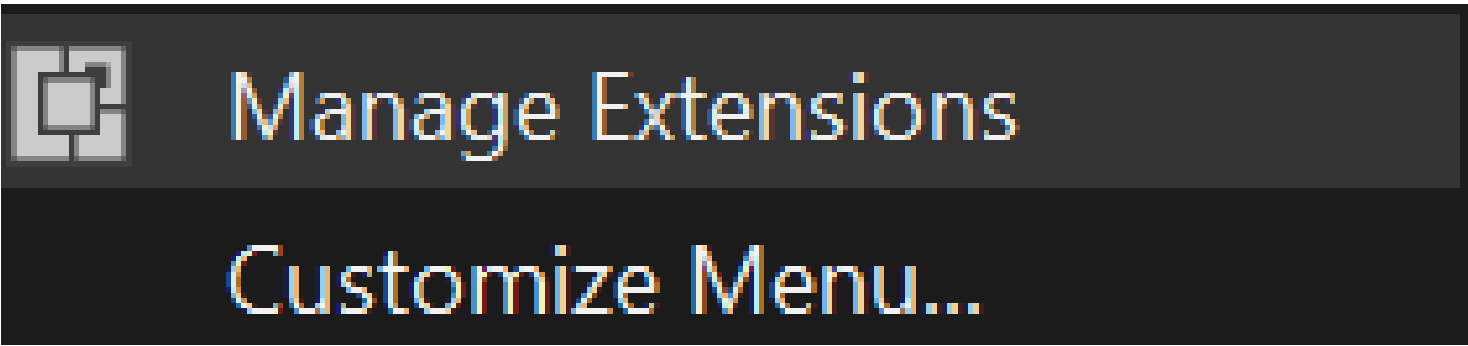
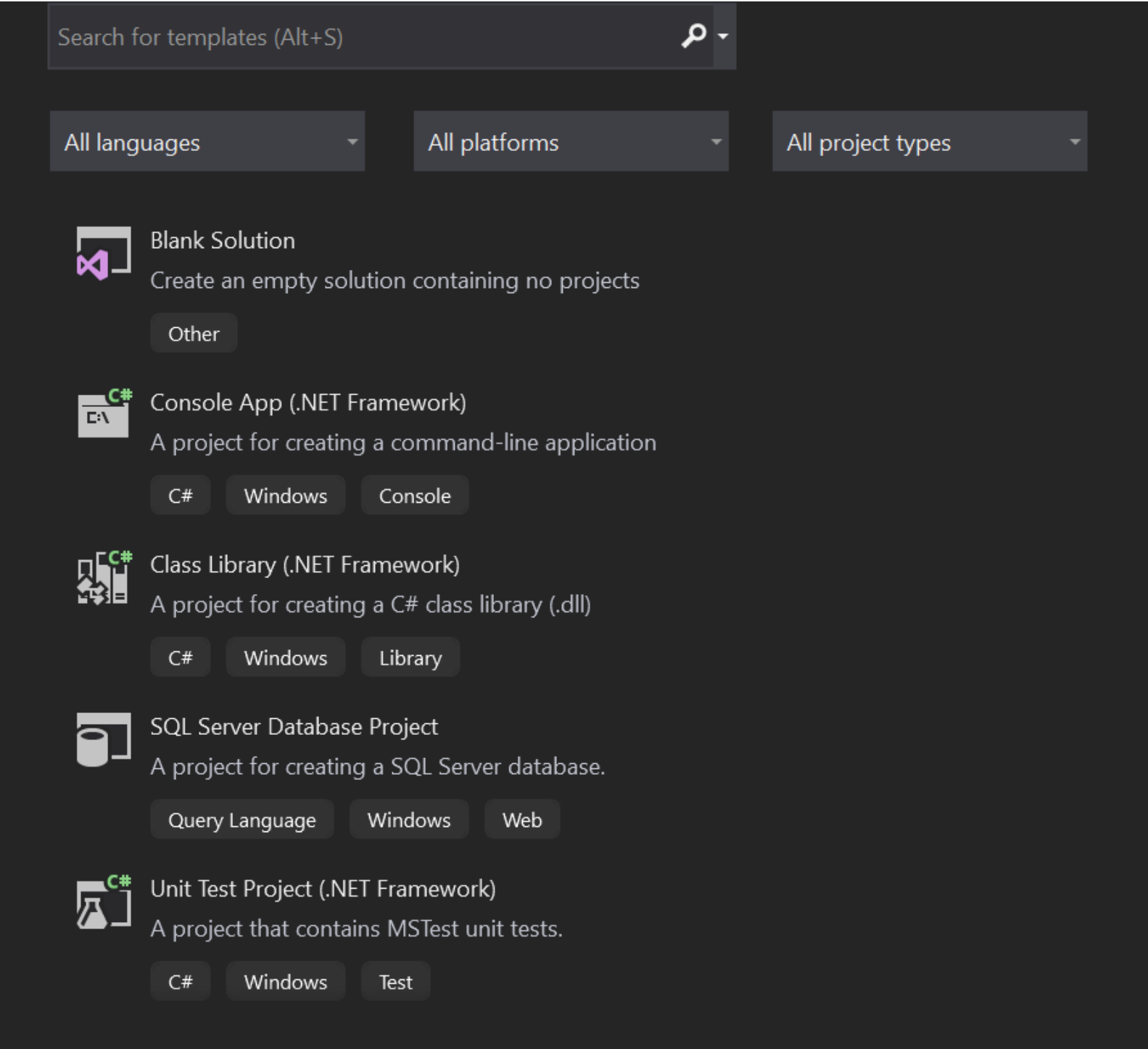
Visual Studio core editor

Data storage and processing

Optional

- ☒ SQL Server Data Tools
- ☐ Azure Data Lake and Stream Analytics Tools
- ☒ .NET Framework 4 – 4.6 development tools
- ☐ F# desktop language support

Individual components



Prerquisito 2

SQL Server Integration Services (SSIS) Tutorial Files

Important! Selecting a language below will dynamically change the complete page content to that language.

Language:

English

[Download](#)



SampleCurrencyData.txt

August 21

Aldo Afranchi



SampleCurre
ncyData.txt

Prerrequisito 3



Express

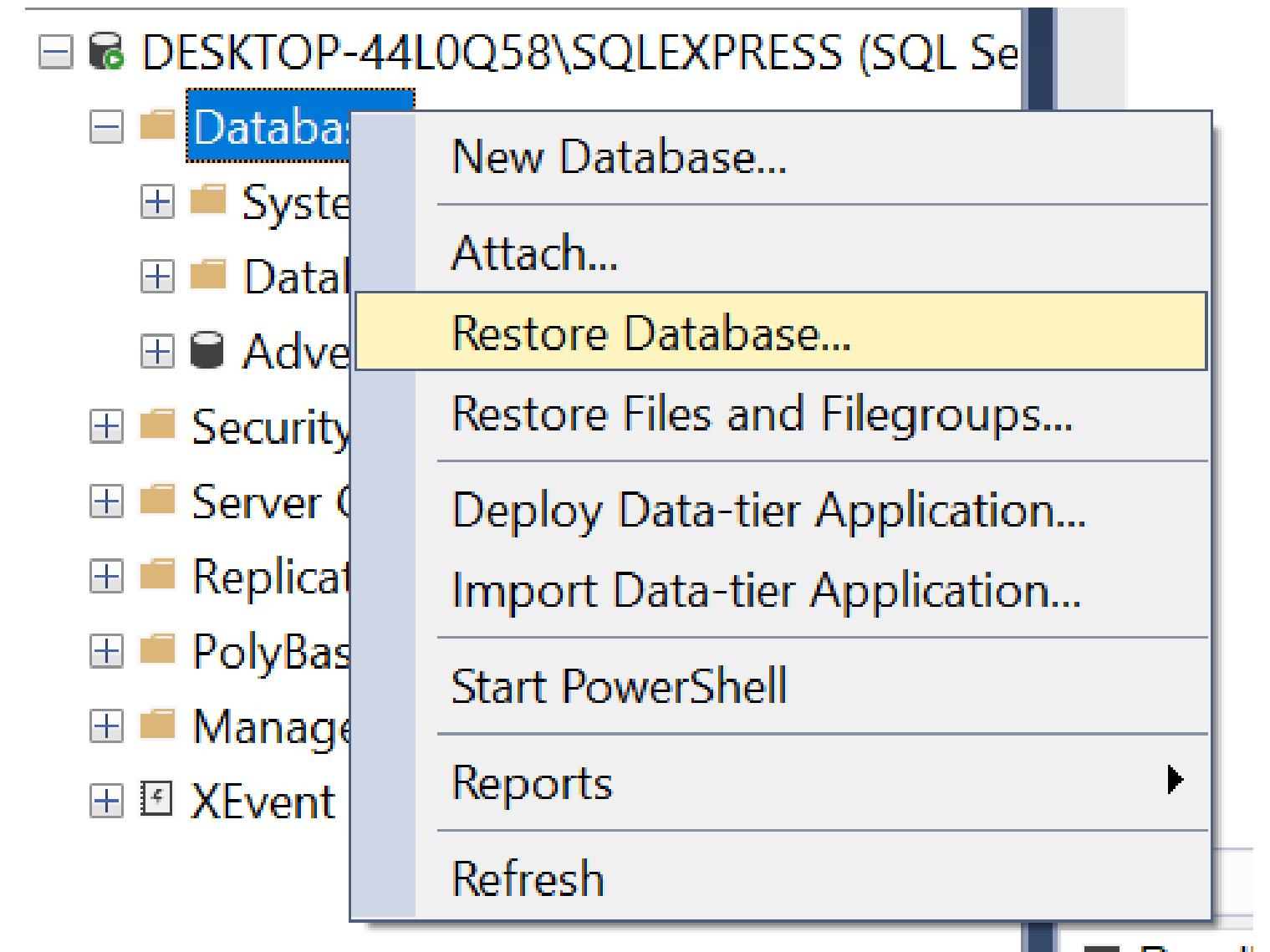
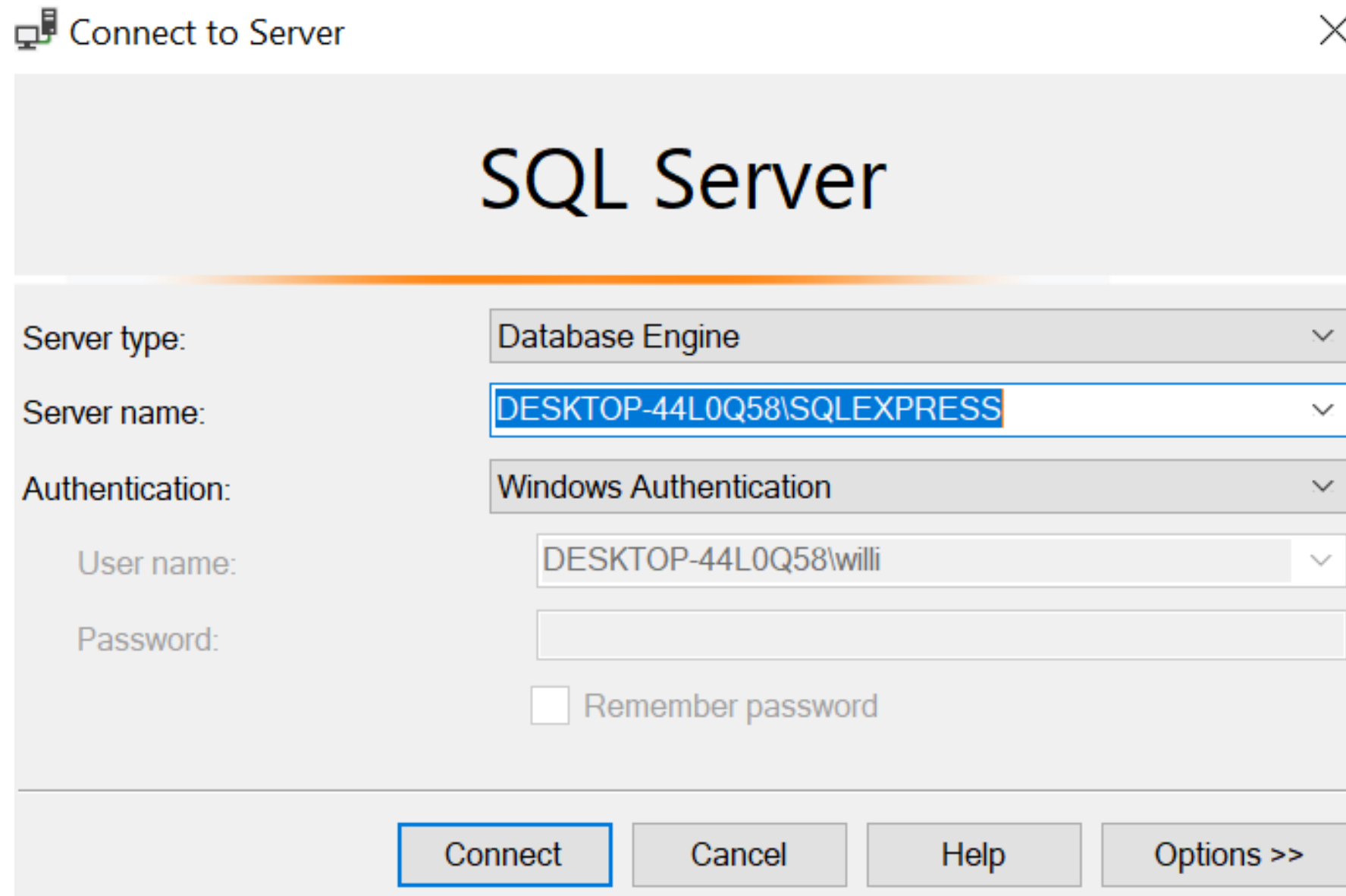
SQL Server 2019 Express is a free edition of SQL Server, ideal for development and production for desktop, web, and small server applications.

[Download now >](#)



Microsoft®
SQL Server®
Management

Prerrequisito 3



Prerrequisito 3

Script ▾ ? Help

Source

☐ Database:

☒ Device: D:\Downloads\Adventure\ ...

Database: AdventureWorksDW2012 ▾

Destination

Database: AdventureWorksDW2012 ▾

Restore to: The last back Timeline...

Restore plan

Backup sets to restore:

Restore	Name	Compo
☒	AdventureWorksDW2012-Full Database Backup	Databa

< >

Verify Backup Media

OK Cancel Help



Lección 1-1

Create a new project

Recent project templates

A list of your recently accessed templates will be displayed here.



All languages

All platforms

All project types

**VB**

Shared Project
Shared Project

Visual Basic

Windows

Desktop



Integration Services Project (Azure-Enabled)
This project may be used for building high performance data integration and workflow solutions that can also be run/debugged on SSIS Platform-as-a-Service (PaaS) in Azure Data Factory.



Integration Services Project
This project may be used for building high performance data integration and workflow solutions that can be run on SSIS catalog, including extraction, transformation, and loading (ETL) operations for data warehousing.



Integration Services Import Project Wizard
A wizard that assists you in creating a new Integration Services (SSIS) project that is based on an existing one. Import from a project deployment file (.ispac extension) or from an Integration Services catalog on an instance of SQL Server.

Lección 1-1

The screenshot displays the Microsoft SQL Server Data Tools (SSDT) interface for designing an SSIS package. The main window is titled "Package.dtsx [Design]" and shows the "Control Flow" tab selected. The design area is currently empty, with a message: "Right-click here to add a new connection manager to the SSIS package." The left sidebar contains the "SSIS Toolbox" with various tasks and containers, and a "Favorites" section. The right sidebar shows the "Solution Explorer" with the project structure, including "SSIS Tutorial (SQL Server 2019)" and "Package.dtsx". Below the Solution Explorer is the "Properties" window for the selected "Package" object.

SSIS Toolbox

- Execute Process Task
- Expression Task
- File System Task
- FTP Task
- Hadoop File System Task
- Hadoop Hive Task
- Hadoop Pig Task
- Script Task
- Send Mail Task
- Web Service Task
- XML Task
- Azure
- Containers
 - For Loop Container
 - Foreach Loop Container
 - Sequence Container
- Other Tasks
- Package Parts
- Favorites**
Provides convenient access to your favorite elements so that they are only a click away.

Package.dtsx [Design]

Control Flow | Data Flow | Parameters | Event Handlers | Package Explorer

100%

Connection Managers

Right-click here to add a new connection manager to the SSIS package.

Output

Show output from: Debug

Solution Explorer

Solution 'EITutorial' (1 of 1 project)

- SSIS Tutorial (SQL Server 2019)
 - Project.params
 - Connection Managers
 - SSIS Packages
 - Package.dtsx
 - Package Parts
 - Control Flow
 - Miscellaneous
 - Linked Azure Resources
 - Azure-SSIS Integration Resources

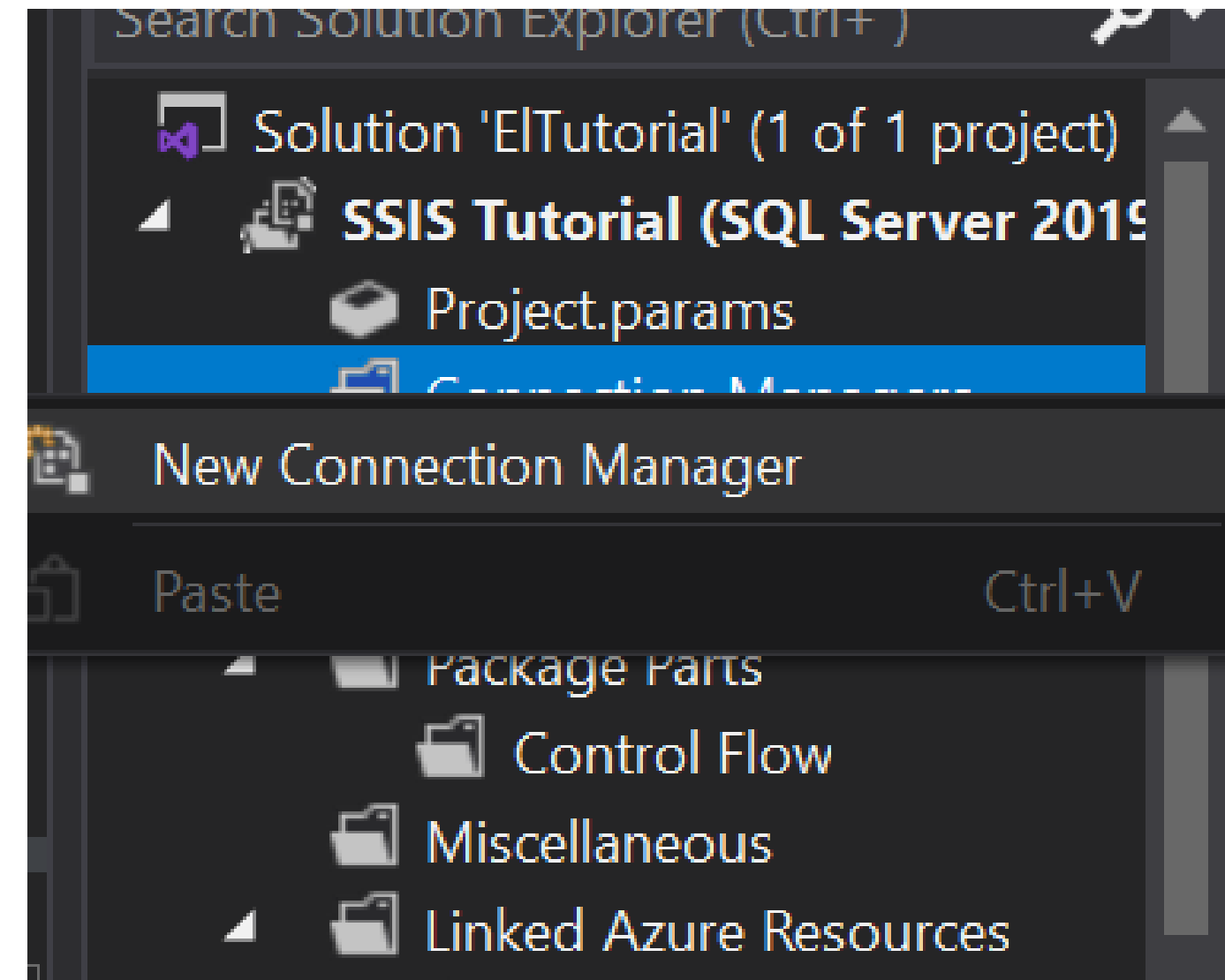
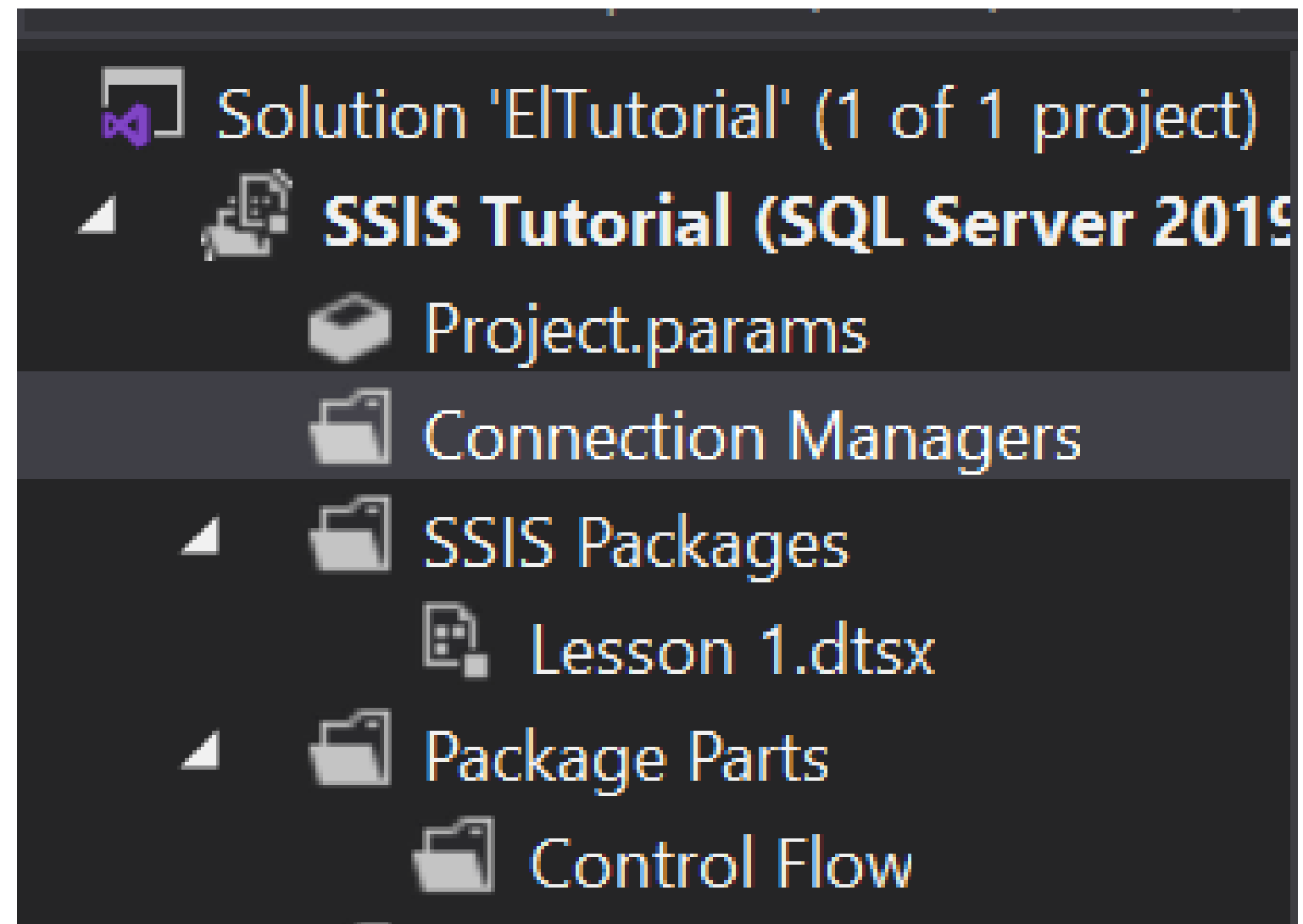
Properties

Package Package

CreatorComputerName	DESKTOP-44L0Q58
CreatorName	DESKTOP-44L0Q58
Description	
ID	{34ADB8F8B-1833-4...
Name	Package
PackageType	DTSDesigner100

Name

Lección 1-2



Lección 1-2

 Add SSIS Connection Manager

Select the type of connection manager to add to the package.

Connection manager type:

Type	Description	File
DQS	Connection manager for DQS server	Micr.
EXCEL	Connection manager for Excel files	D:\P.
FILE	Connection manager for files	D:\P.
FLATFILE	Connection manager for flat files	D:\P.
FTP	Connection manager for FTP connections	D:\P.
Hadoop	Connection manager for Hadoop	Micr.

Flat File Connection Manager Editor

Connection manager name: Sample Flat File Source Data.

Description:

General
Columns
Advanced
Preview

Select a file and specify the file properties and the file format.

File name: D:\Desktop\Creating a Simple ETL Package Browse...

Locale: English (United States) ☐ Unicode

Code page: 1252 (ANSI - Latin I)

Format: Delimited

Text qualifier: <none>

Header row delimiter: {CR}{LF}

Header rows to skip: 0

☐ Column names in the first data row

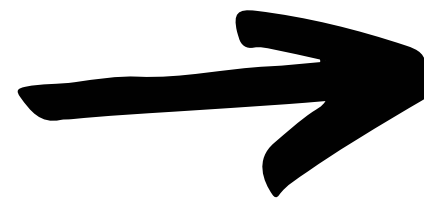
 Columns are not defined for this connection manager.

OK Cancel Help

Lección 1-2

Configure the properties of each column.

Column 0
Column 1
Column 2
Column 3

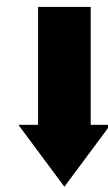


Configure the properties of each column.

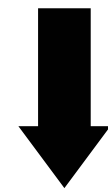
AverageRate	N
CurrencyID	N
CurrencyDate	C
EndOfDayRate	C
	Ir
	D
	D
	D

Lección 1-2

AverageRate
CurrencyID
CurrencyDate



DataScale	0
DataType	string [DT_STR]
OutputColumnW	3



Unicode string [DT_WSTR]

CurrencyID
CurrencyDate
EndOfDayRate



DataScale	0
DataType	date [DT_DATE]
OutputColumnW	0



database date [DT_DBDATE]

Lección 1-3

Connection manager type:

Type	Description	File .
Hadoop	Connection manager for Hadoop	Micr.
HTTP	Connection manager for HTTP connections	D:\P.
MSMQ	Connection manager for the Message Queue task	Micr.
MSOLAP100	Connection manager for Analysis Services conn...	D:\P.
MULTIFILE	Connection manager for multiple files	D:\P.
MULTIFLATFILE	Connection manager for multiple flat files	D:\P.
ODATA	Connection Manager for OData services	Micr.
ODBC	Connection manager for ODBC connections	D:\P.
OLEDB	Connection manager for OLE DB connections	D:\P.
SMOServer	Connection manager for SQL Server transfer tas...	Micr.
SMTP	Connection manager for the Send Mail task	Micr.
SQLMOBILE	Connection manager for SQL Server Compact c...	D:\P.
WMI	Connection manager for the WMI tasks	Micr.

Data connections:

Data connection properties:

Property	Value
----------	-------

New... Delete

OK Cancel

Lección 1-3

Provider: Native OLE DB\SQL Server Native Client 11.0 ▼

Connection

All

Server name:
DESKTOP-44L0Q58\SQLEXPRESS ▼ Refresh

Log on to the server

Authentication: Windows Authentication ▼

User name:

Password:

☐ Save my password

Connect to a database

☒ Select or enter a database name:
AdventureWorksDW2012 ▼

☐ Attach a database file:
 Browse...
Logical name:

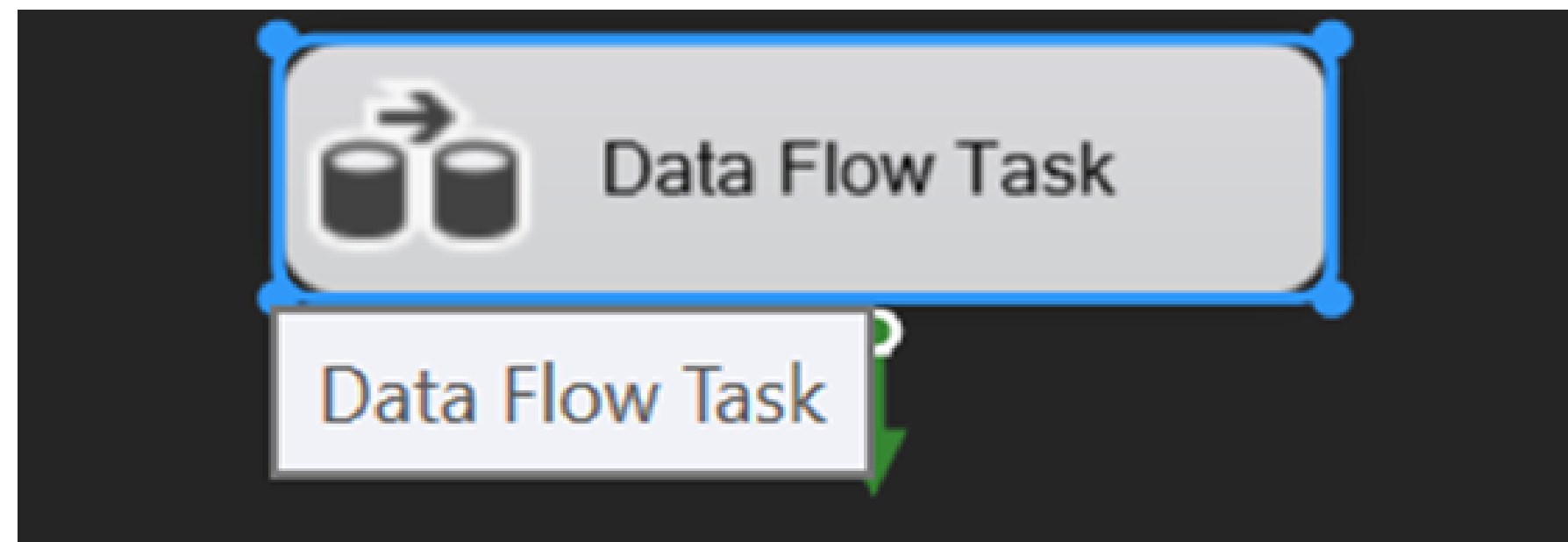
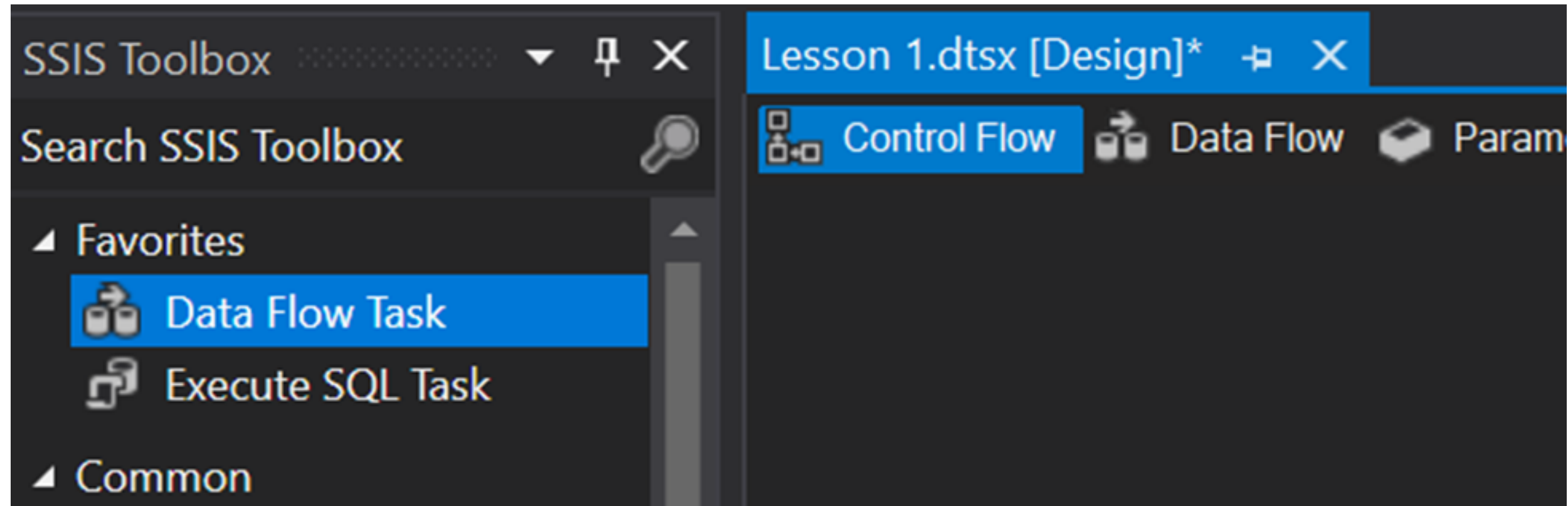
Test Connection OK Cancel Help

Connection Manager

Test connection succeeded.

 Copy message

Lección 1-4



Lección 1-4



Properties

Alt+Enter

ForceExecutionResult

None

HasExpressions

False

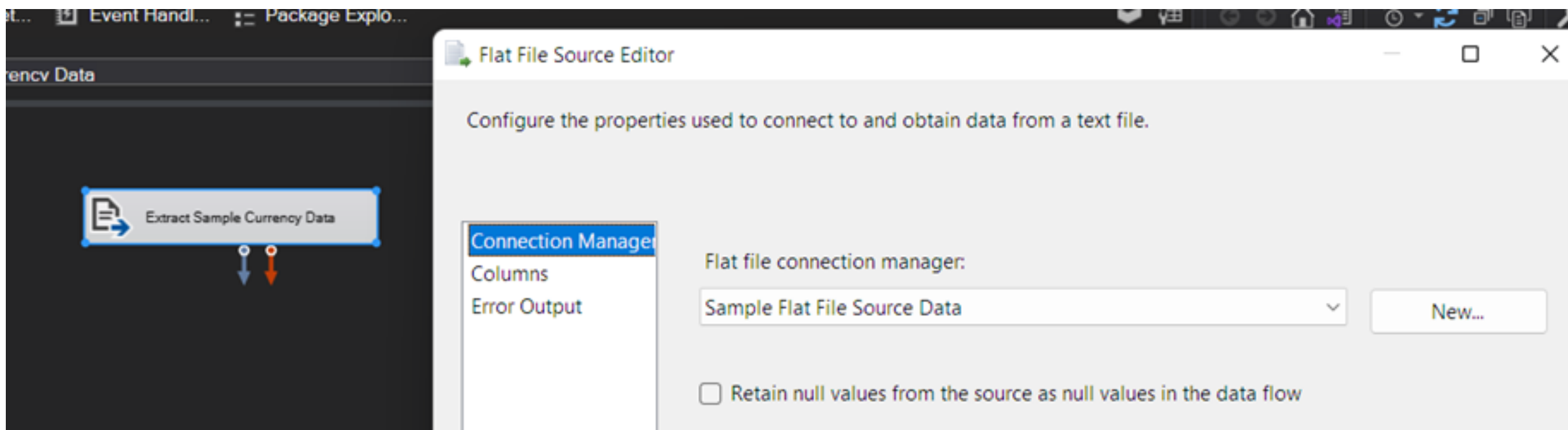
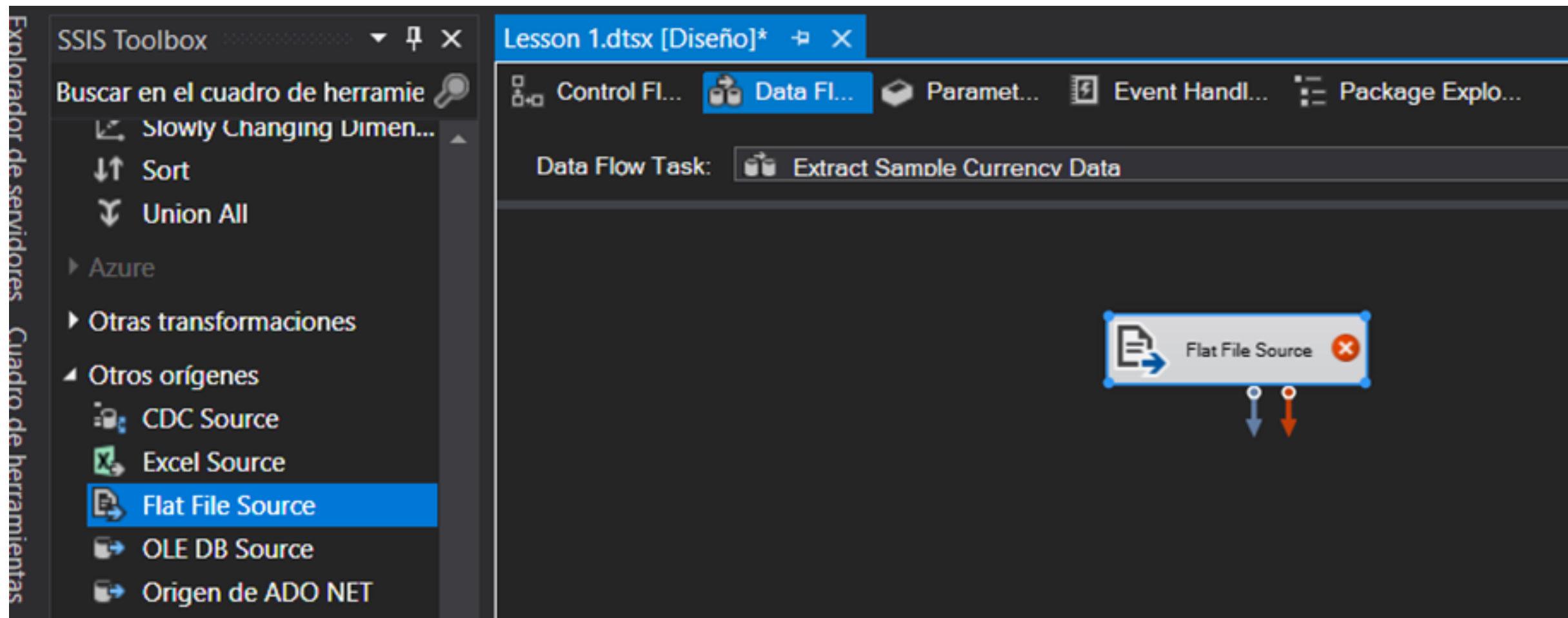
LocaleID

English (United States)

LoggingMode

UseParentSetting

Lección 1-5



Lección 1-5

Configure the properties used to connect to and obtain data from a text file.

Connection Manager
Columns
Error Output

Available External Colu...

☒ Name

☒ AverageRate

☒ CurrencyID

☒ CurrencyDate

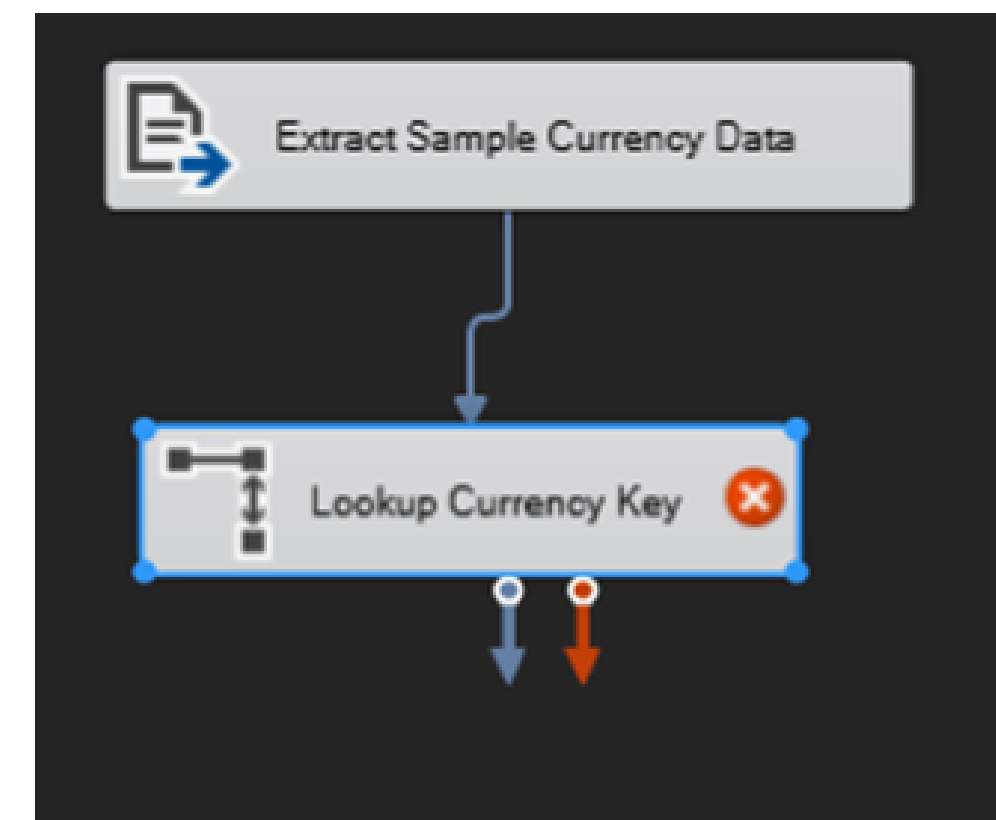
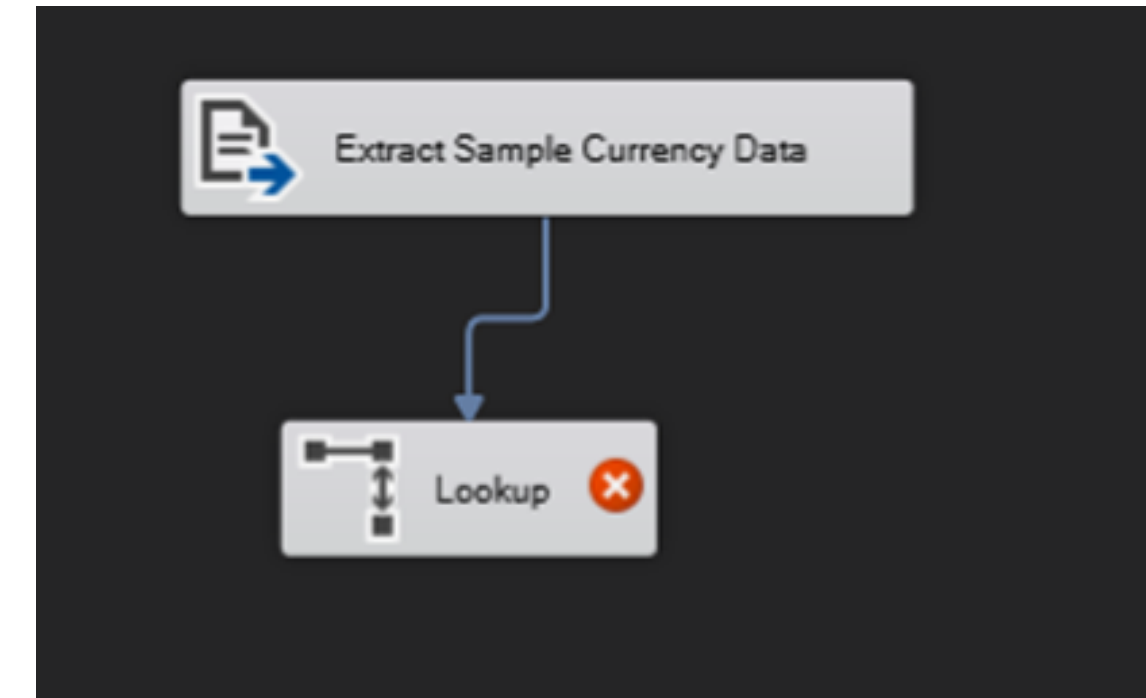
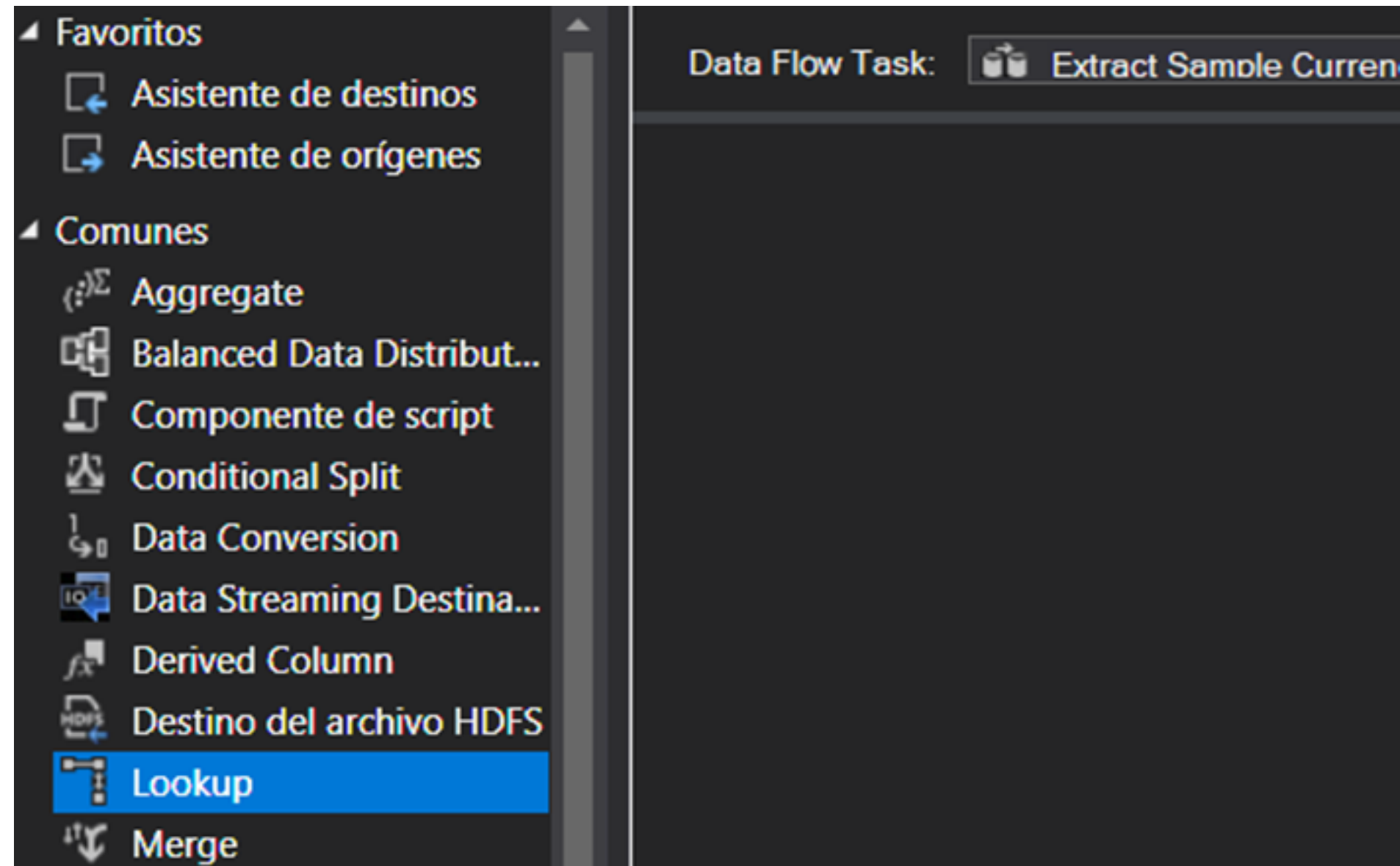
External Column	Output Column
AverageRate	AverageRate
CurrencyID	CurrencyID
CurrencyDate	CurrencyDate
EndOfDayRate	EndOfDayRate

Propiedades

Extract Sample Currency Data Data Flow Component

ComponentClassID	{FC58D298-4FC7-413C-9E80-D}
ContactInfo	Flat File Source;Microsoft Corp
Description	Flat File Source
ID	1
IdentificationString	Extract Sample Currency Data
IsDefaultLocale	False
LocaleID	Inglés (Estados Unidos)
Name	Extract Sample Currency Dat
PipelineVersion	0

Lección 1-6



Lección 1-6

This screenshot shows the 'General' tab of the Lookup Transformation Editor. On the left, a sidebar lists the tabs: General, Connection, Columns, Advanced, and Error Output. The main area contains three sections: 'Cache mode' with three radio buttons (Full cache is selected), 'Connection type' with two radio buttons (OLE DB connection manager is selected), and a dropdown menu for 'Specify how to handle rows with no matching entries' set to 'Fail component'.

General
Connection
Columns
Advanced
Error Output

Cache mode

☒ Full cache
☐ Partial cache
☐ No cache

Connection type

☐ Cache connection manager
☒ OLE DB connection manager

Specify how to handle rows with no matching entries

Fail component

This screenshot shows the 'Connection' tab of the Lookup Transformation Editor. The title bar reads 'Lookup Transformation Editor'. Below the title bar, a description states: 'This transform enables the performance of simple equi-joins between the input and a reference data set.' The left sidebar is the same as the previous image, with 'Connection' now selected. The main area has a text prompt: 'Specify a data source to use. You can select a table in a data source view, a table in a database connection, or the results of an SQL query.' There are three options: 'OLE DB connection manager' (selected), 'Use a table or a view:', and 'Use results of an SQL query:'. The 'OLE DB connection manager' option has a dropdown menu showing 'LAPTOP-1096KIB9_SQLEXPRESS.AdventureWorksDW2012' and a 'New...' button. The 'Use results of an SQL query:' option has a text area containing an SQL query and buttons for 'Build Query...', 'Browse...', 'Parse Query', and 'Preview...'. At the bottom are 'OK', 'Cancel', and 'Help' buttons.

Lookup Transformation Editor

This transform enables the performance of simple equi-joins between the input and a reference data set.

General
Connection
Columns
Advanced
Error Output

Specify a data source to use. You can select a table in a data source view, a table in a database connection, or the results of an SQL query.

OLE DB connection manager:

LAPTOP-1096KIB9_SQLEXPRESS.AdventureWorksDW2012 New...

☐ Use a table or a view:

New...

☒ Use results of an SQL query:

```
SELECT * FROM [dbo].[DimCurrency]WHERE  
[CurrencyAlternateKey]IN ('ARS', 'AUD', 'BRL', 'CAD', 'CNY',  
'DEM', 'EUR', 'FRF', 'GBP', 'JPY', 'MXN', 'SAR', 'USD', 'VEB')
```

Build Query...
Browse...
Parse Query

Preview...

OK Cancel Help

Lección 1-6

Preview Query Results

Query result (up to the first 200 rows):

CurrencyKey	CurrencyAlternateKey	CurrencyName
3	ARS	Argentine Peso
6	AUD	Australian Dollar
16	BRL	Brazilian Real
19	CAD	Canadian Dollar
103	CNY	Yuan Renminbi
29	DEM	Deutsche Mark
36	EUR	EURO
39	FRF	French Franc
98	GBP	United Kingdom Pou...
102	JPY	Yen
61	MXN	Mexican Peso
85	SAR	Saudi Riyal
100	USD	US Dollar
14	VEB	Bolivar

Close

General
Connection
Columns
Advanced
Error Output

Available Input Co...

Name	
AverageRate	
CurrencyID	
CurrencyDate	
EndOfDayRate	

Available Lookup Columns

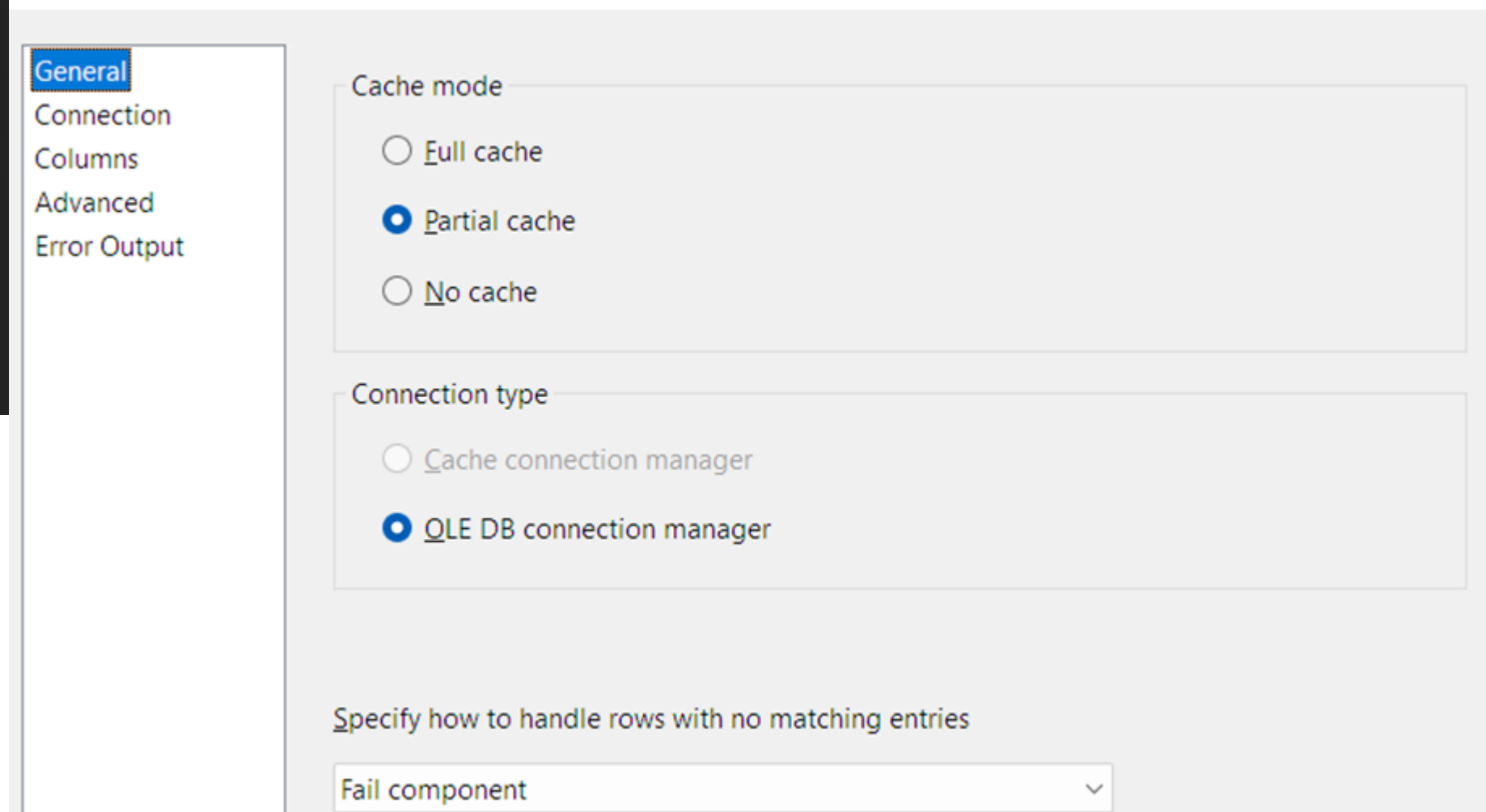
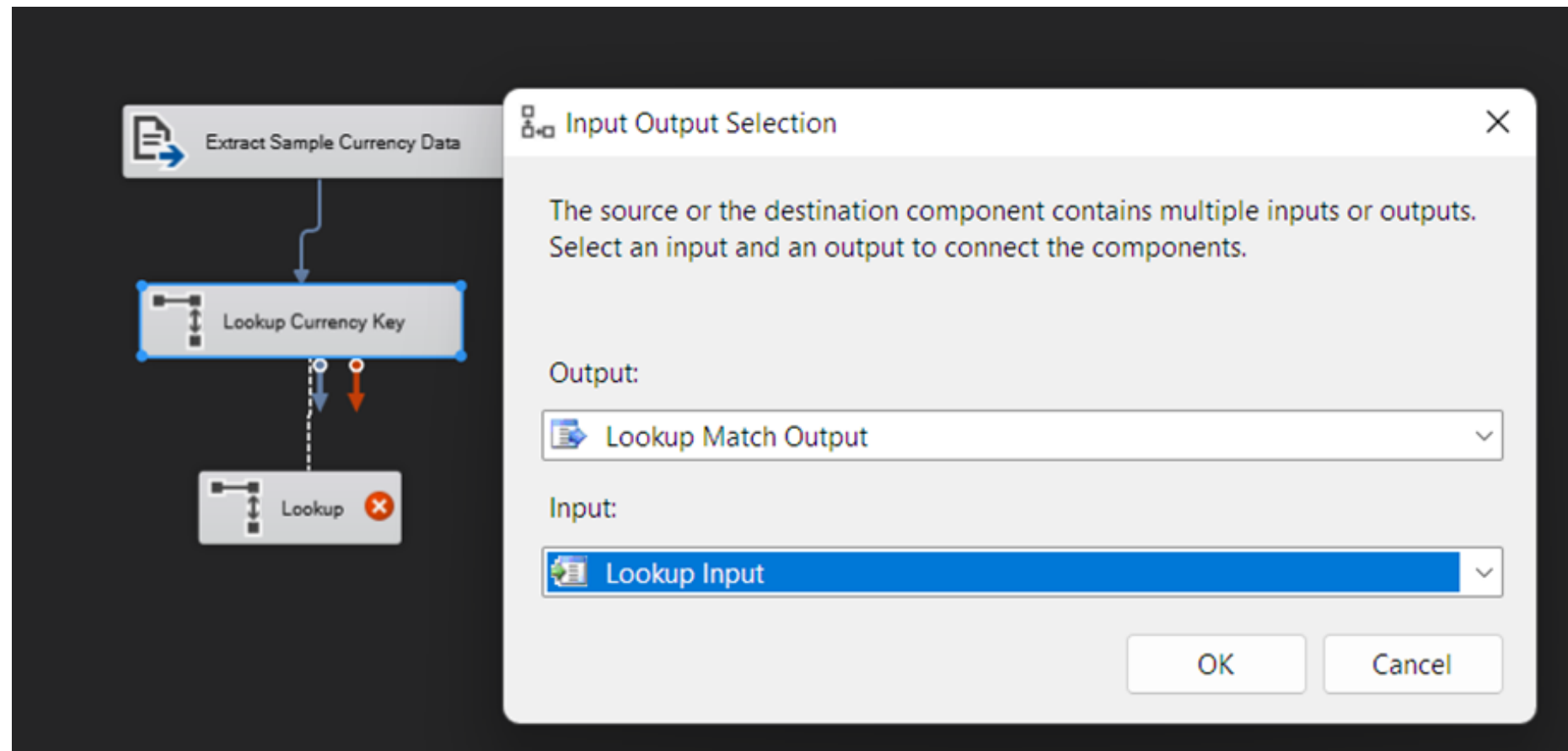
☐ Name	I...
☒ CurrencyKey	
☐ CurrencyAlternateKey	
☐ CurrencyName	

Lookup Column	Lookup Operation	Output Alias
CurrencyKey	<add as new column>	CurrencyKey

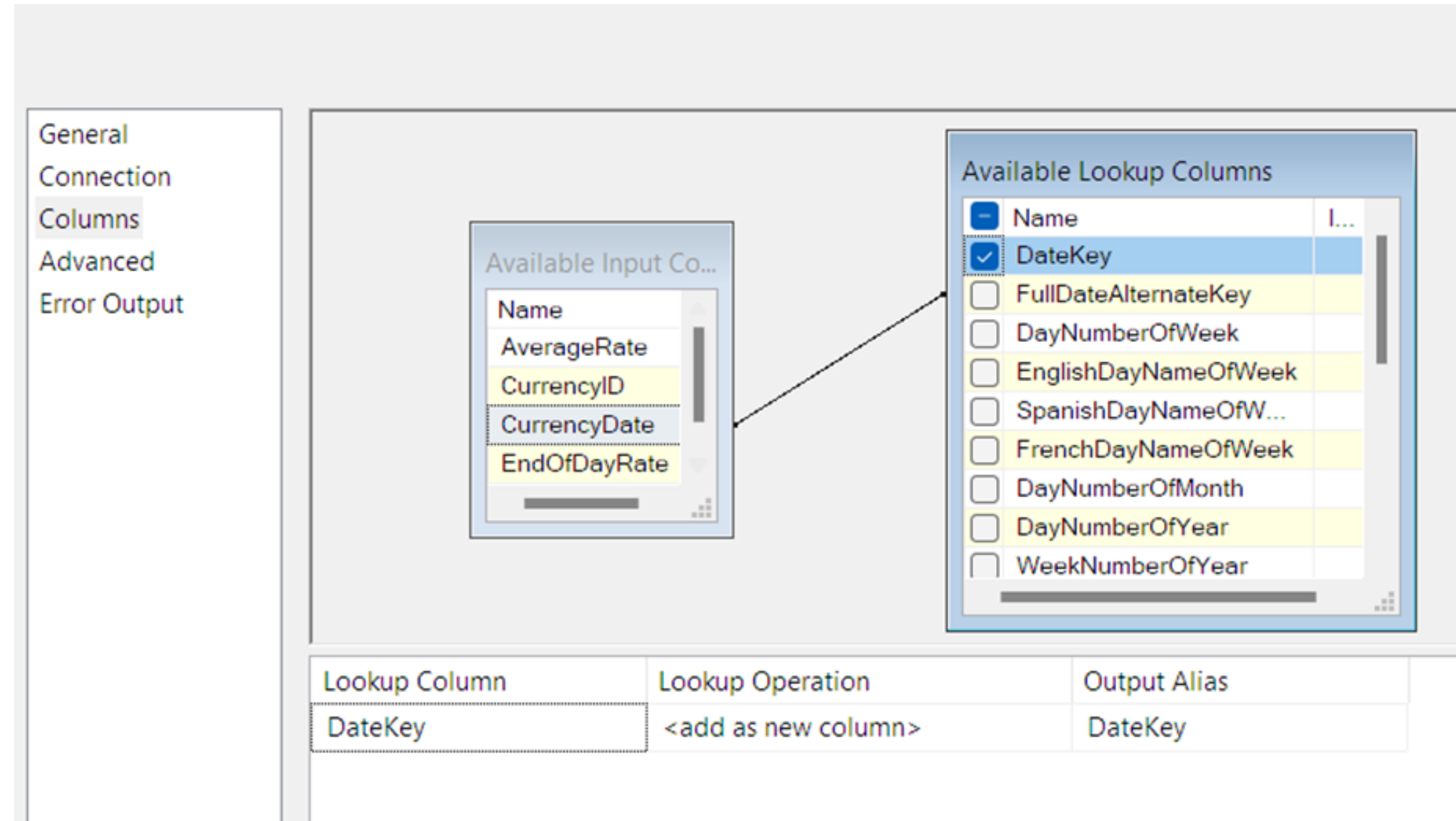
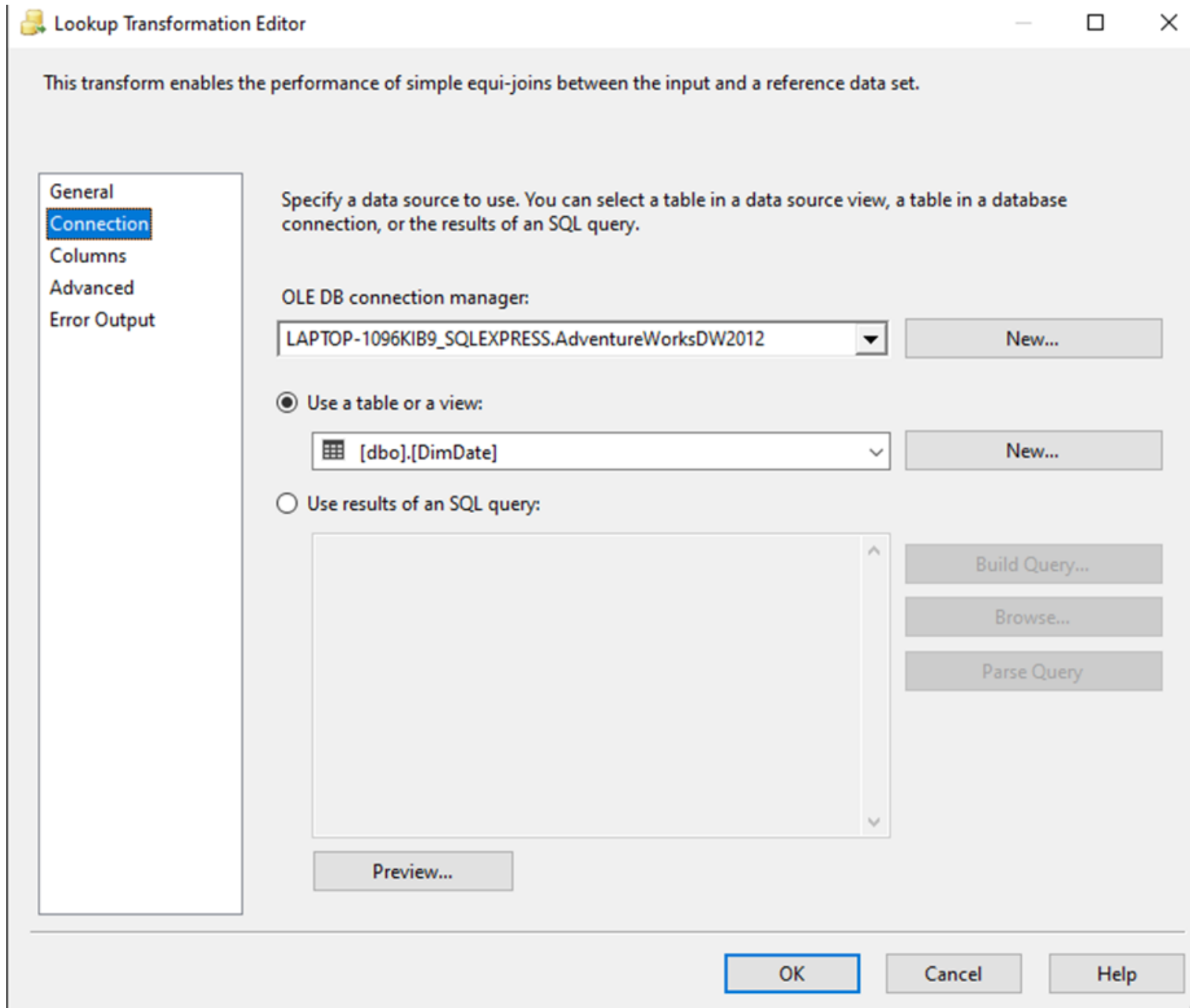
Custom Properties

CacheType	Full
ConnectionType	OleDb
DefaultCodePage	1252
NoMatchBehavior	Treat rows with no ma

Lección 1-6



Lección 1-6



Lección 1-6

This transform enables the performance of simple equi-joins between the input and a reference data set.

General

Connection

Columns

Advanced

Error Output

Partial caching

Cache size (32-bit):

25

MB

Cache size (64-bit):

25

MB

☐ Enable cache for rows with no matching entries

Allocation from cache

0

Percent (%)

Custom query

☐ Modify the SQL statement

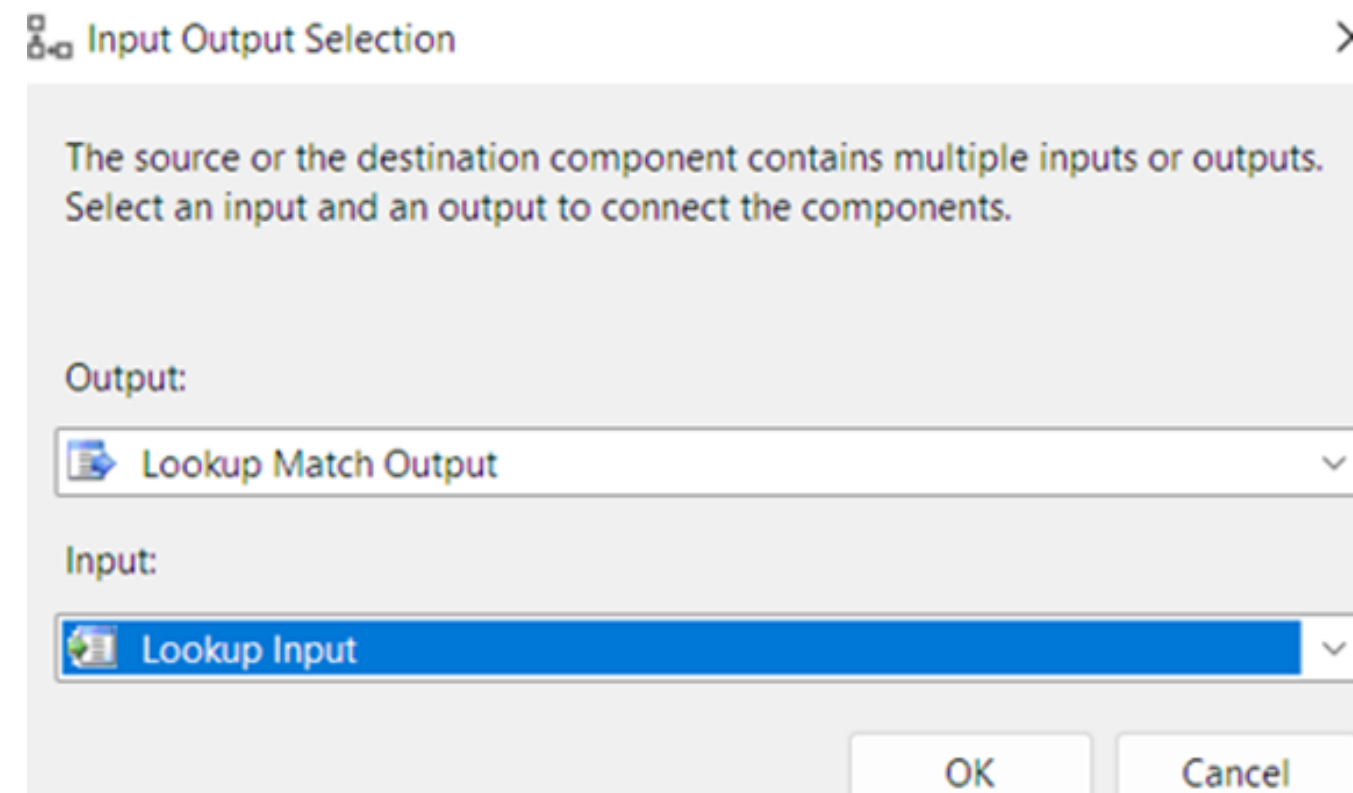
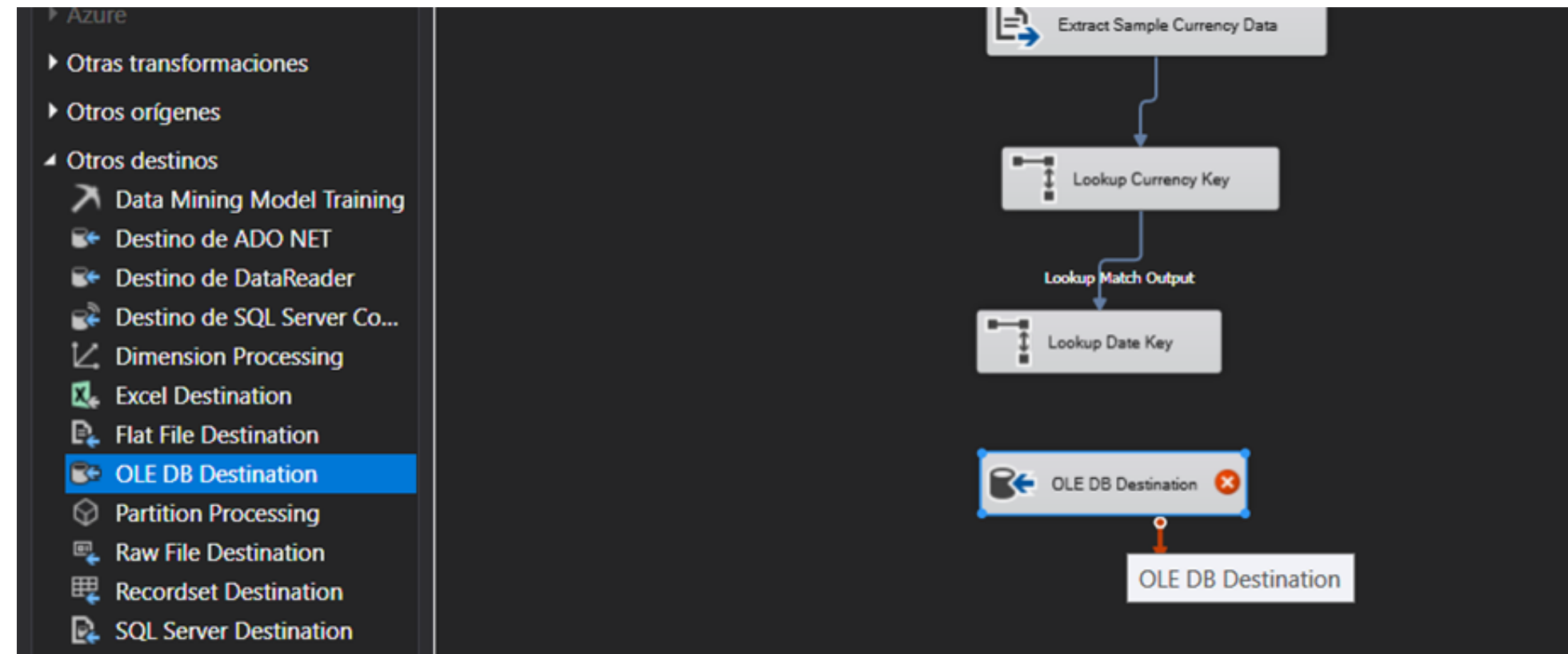
select * from (select * from [dbo].[DimDate]) [refTable]
where [refTable].[FullDateAlternateKey] = ?

Parameters...

Approximate cached row size (bytes): 42

Custom Properties	
CacheType	Full
ConnectionType	OLEDB
DefaultCodePage	1252
NoMatchBehavior	Treat rows with no ma

Lección 1-7



Lección 1-7

OLE DB Destination Editor

Configure the properties used to insert data into a relational database using an OLE DB provider.

Connection Manager
Mappings
Error Output

Specify an OLE DB connection manager, a data source, or a data source view, and select the data access mode. If using the SQL command access mode, specify the SQL command either by typing the query or by using Query Builder. For fast-load data access, set the table update options.

OLE DB connection manager:
LAPTOP-1096KIB9_SQLEXPRESS.AdventureWorksDW2012 New...

Data access mode:
Table or view - fast load

Name of the table or the view:
[NewFactCurrencyRate.] New...

☐ Keep identity ☒ Table lock
☐ Keep nulls ☒ Check constraints

Rows per batch:
Maximum insert commit size: 2147483647

View Existing

OK Cancel Help

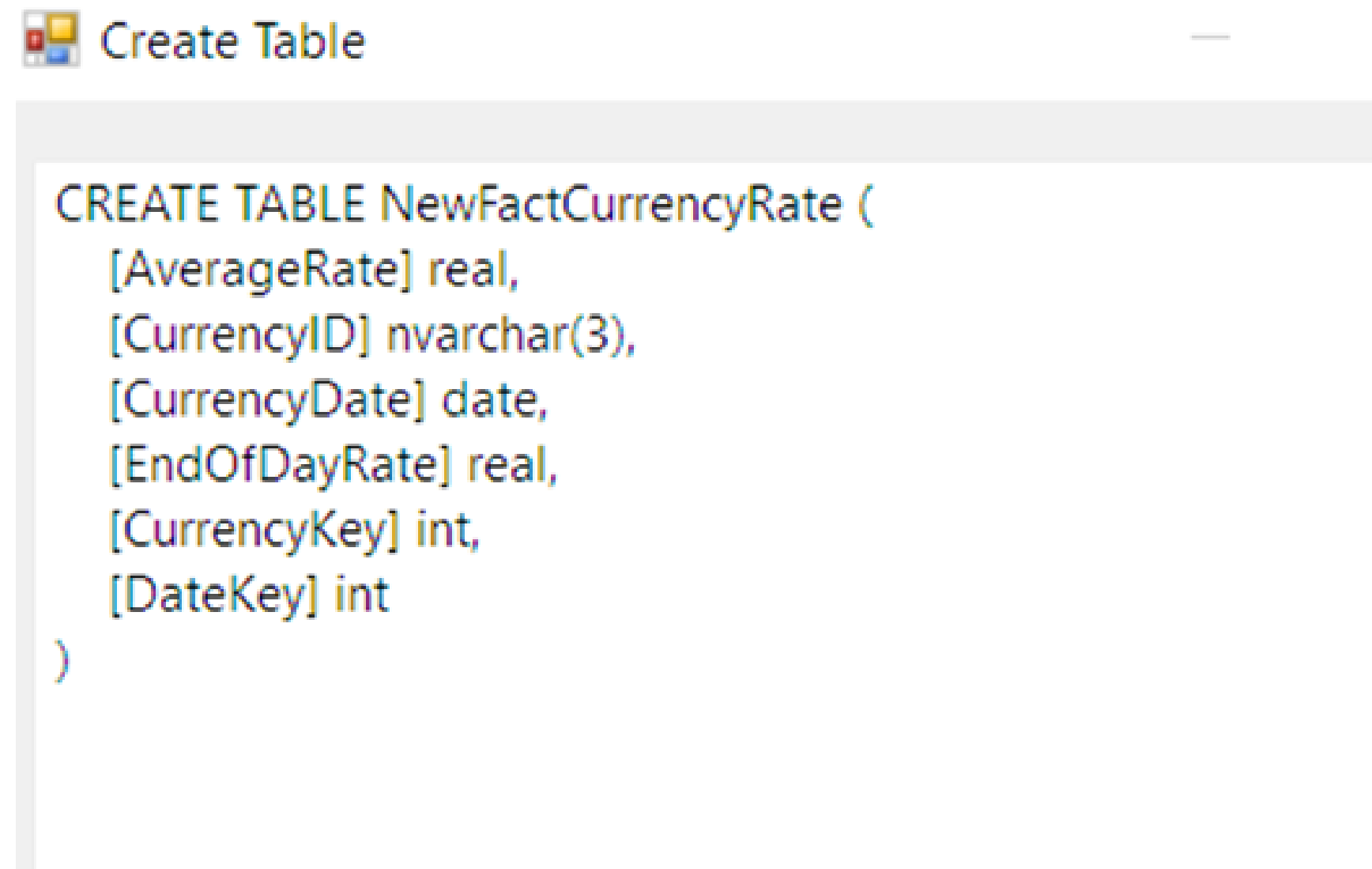
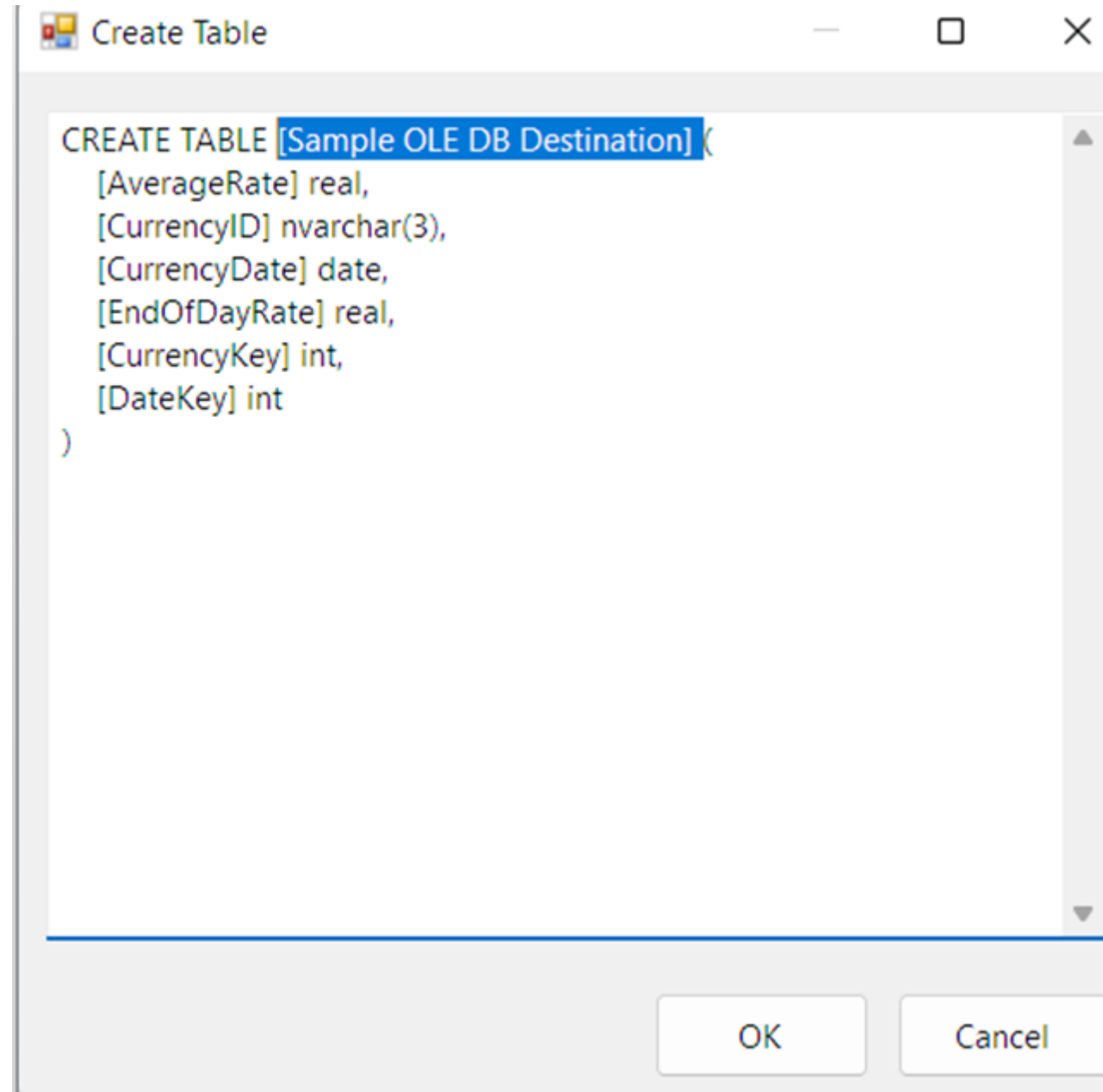
[dbo].[FactSurveyResponse]
[dbo].[NewFactCurrencyRate]
[dbo].[ProspectiveRiver]

table or view - fast load

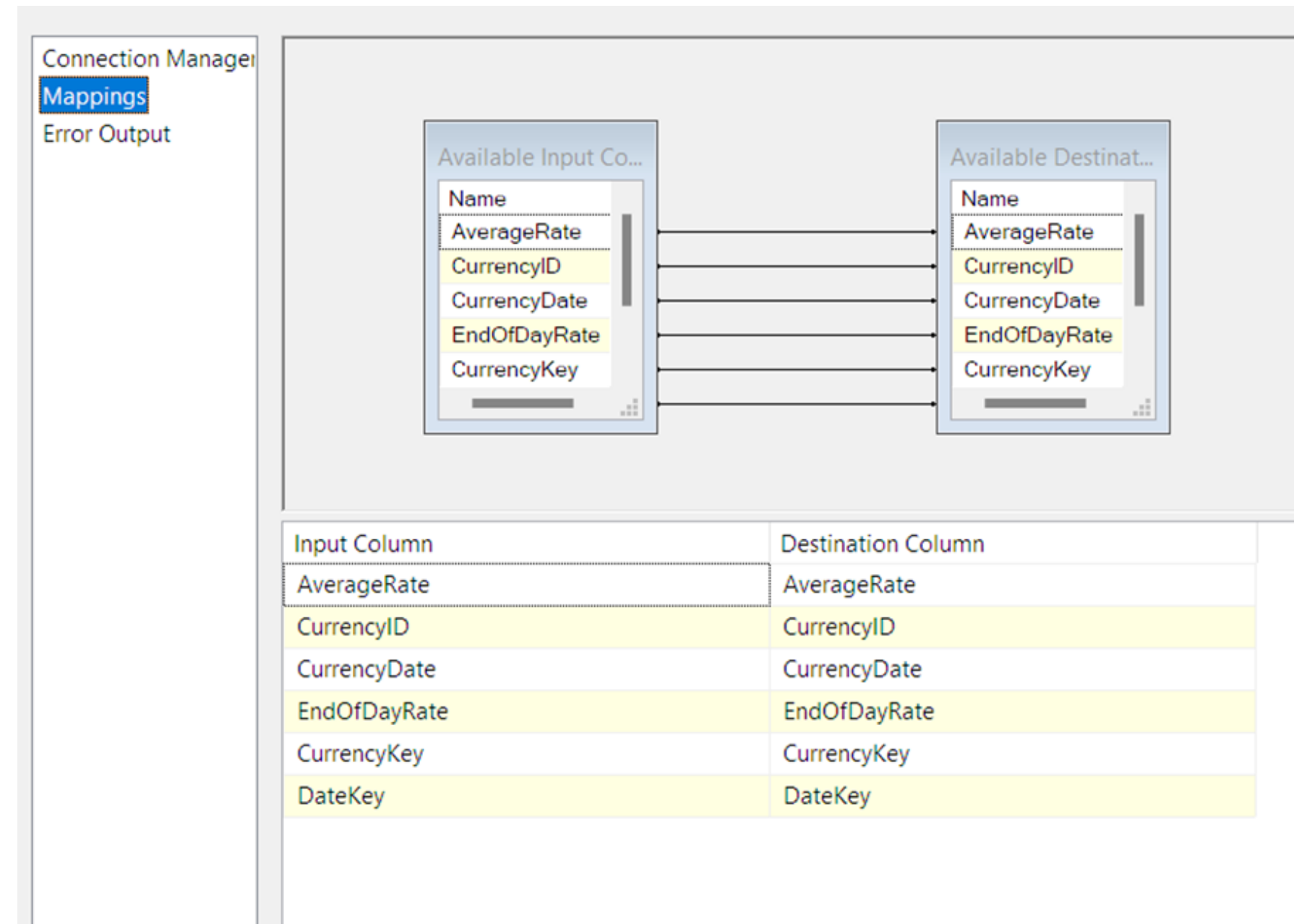
Name of the table or the view:
[dbo].[FactCurrencyRate] New...

☐ Keep identity ☒ Table lock

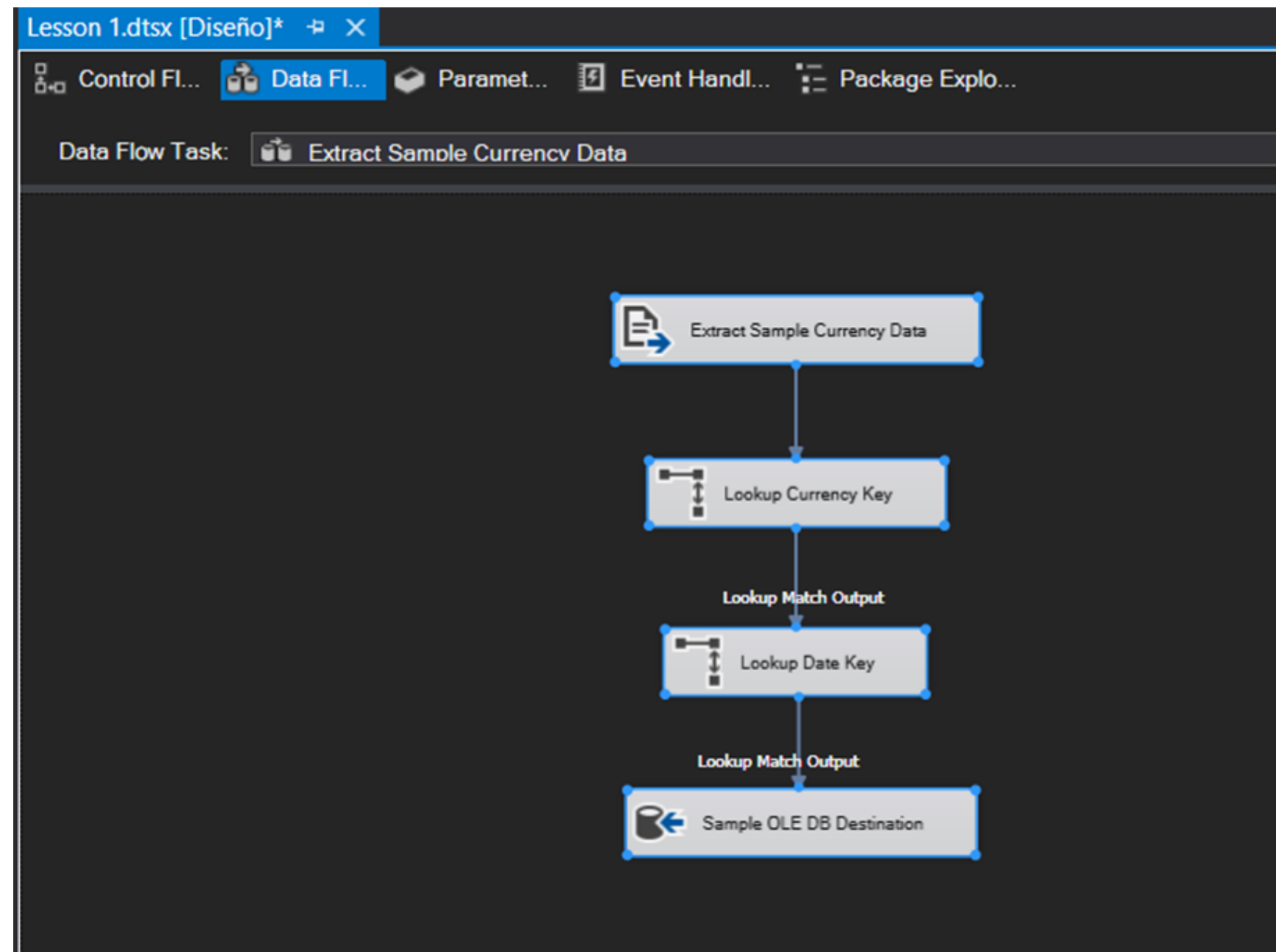
Lección 1-7



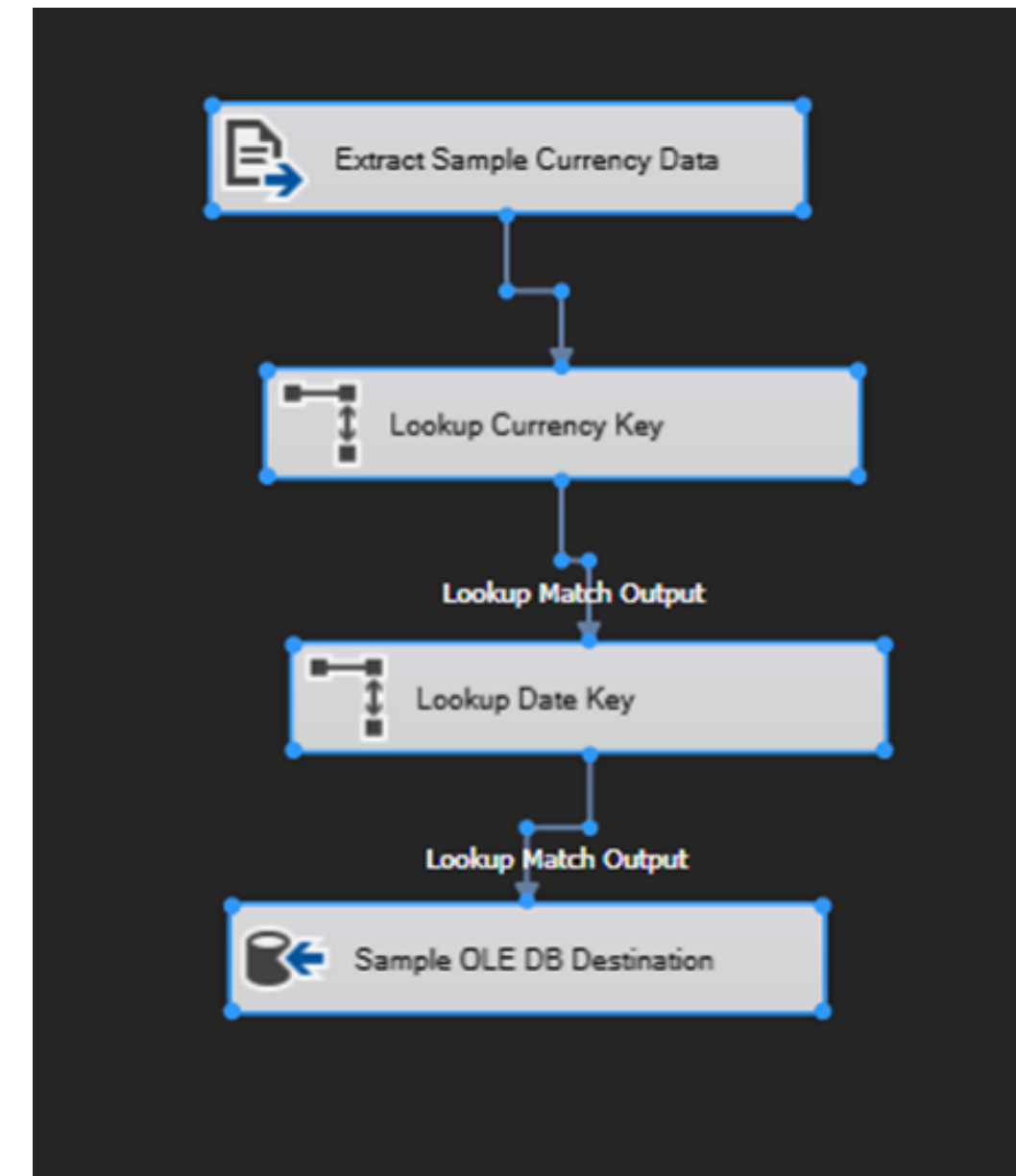
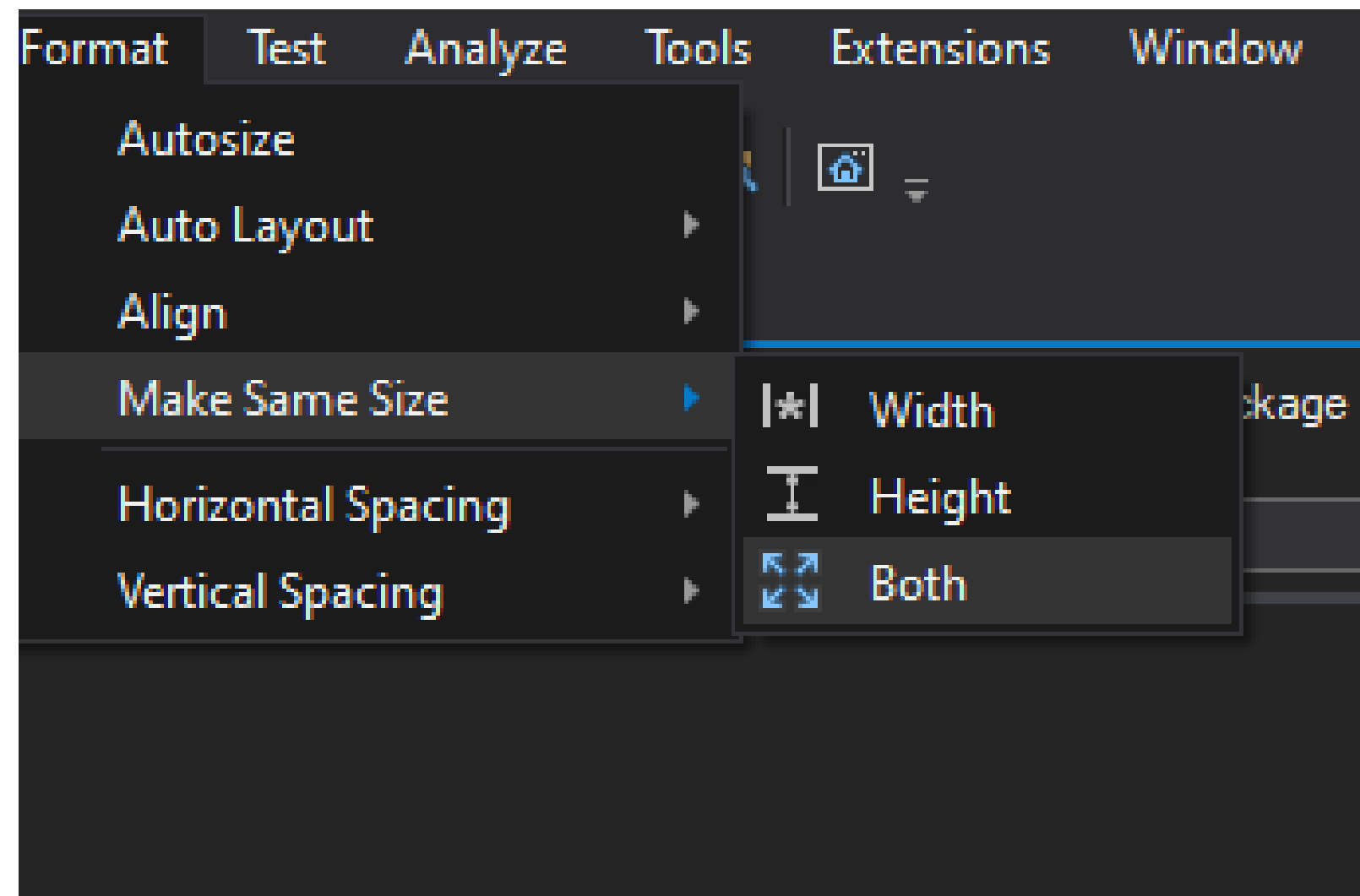
Lección 1-7



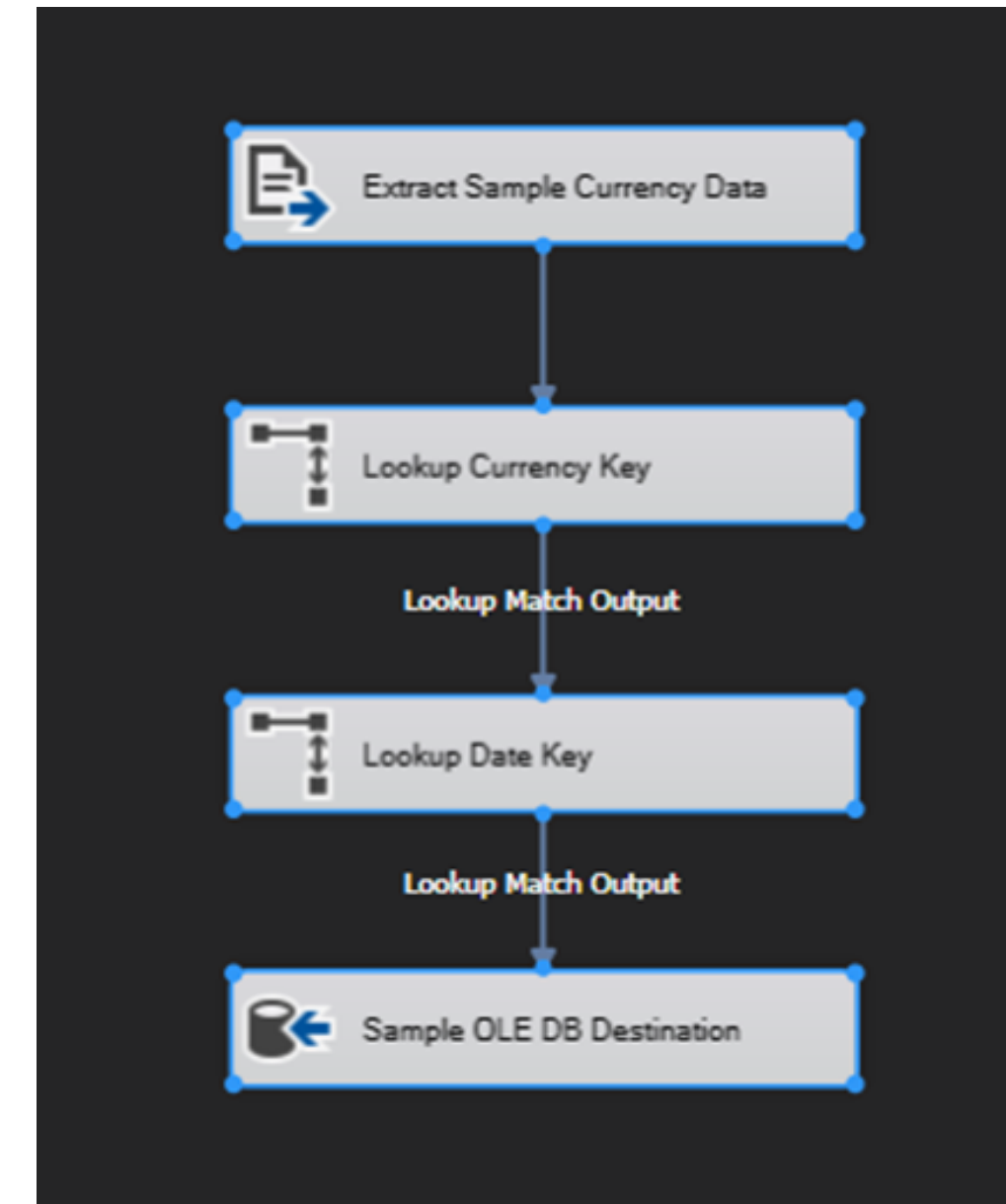
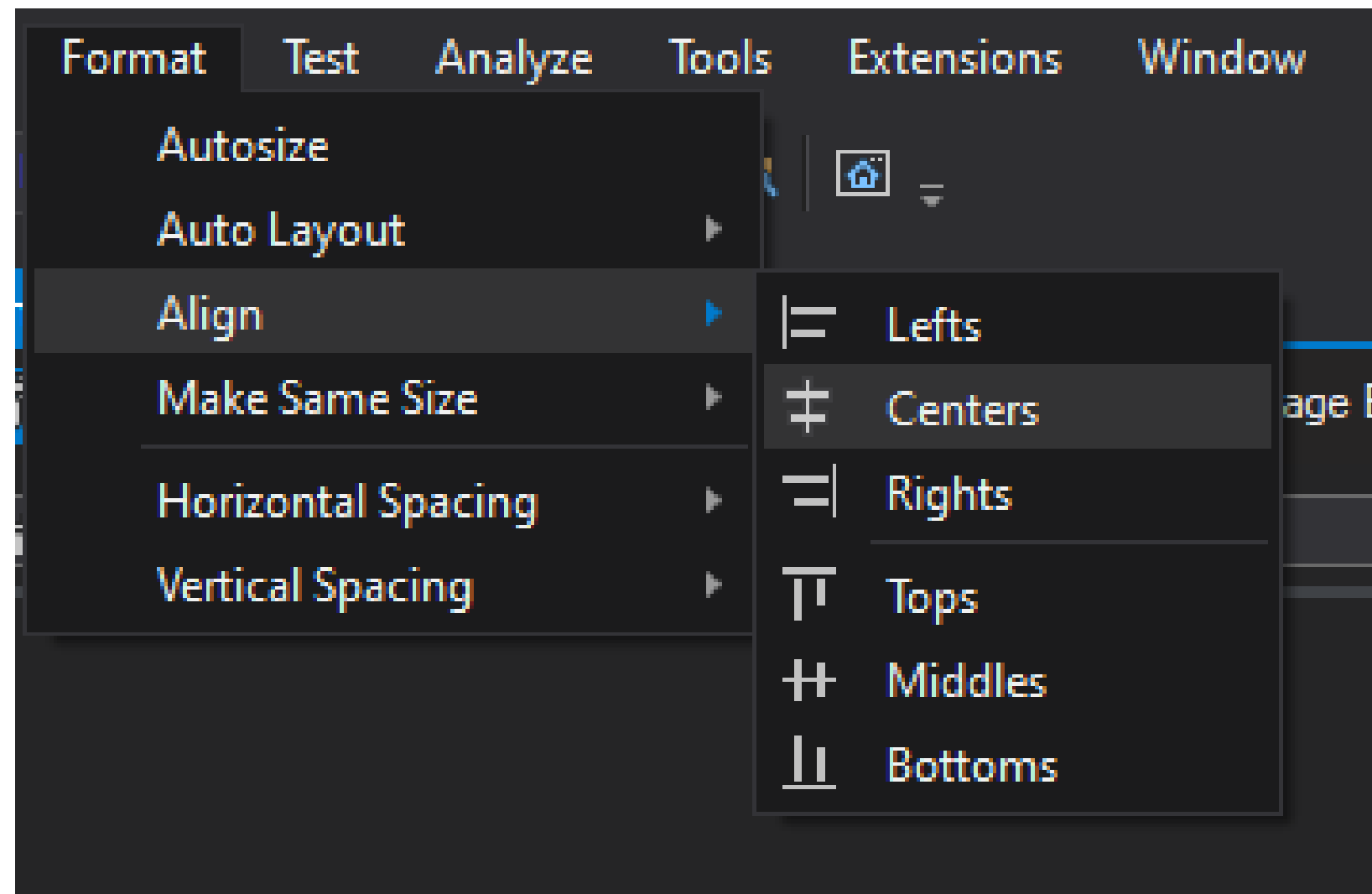
Lección 1-8



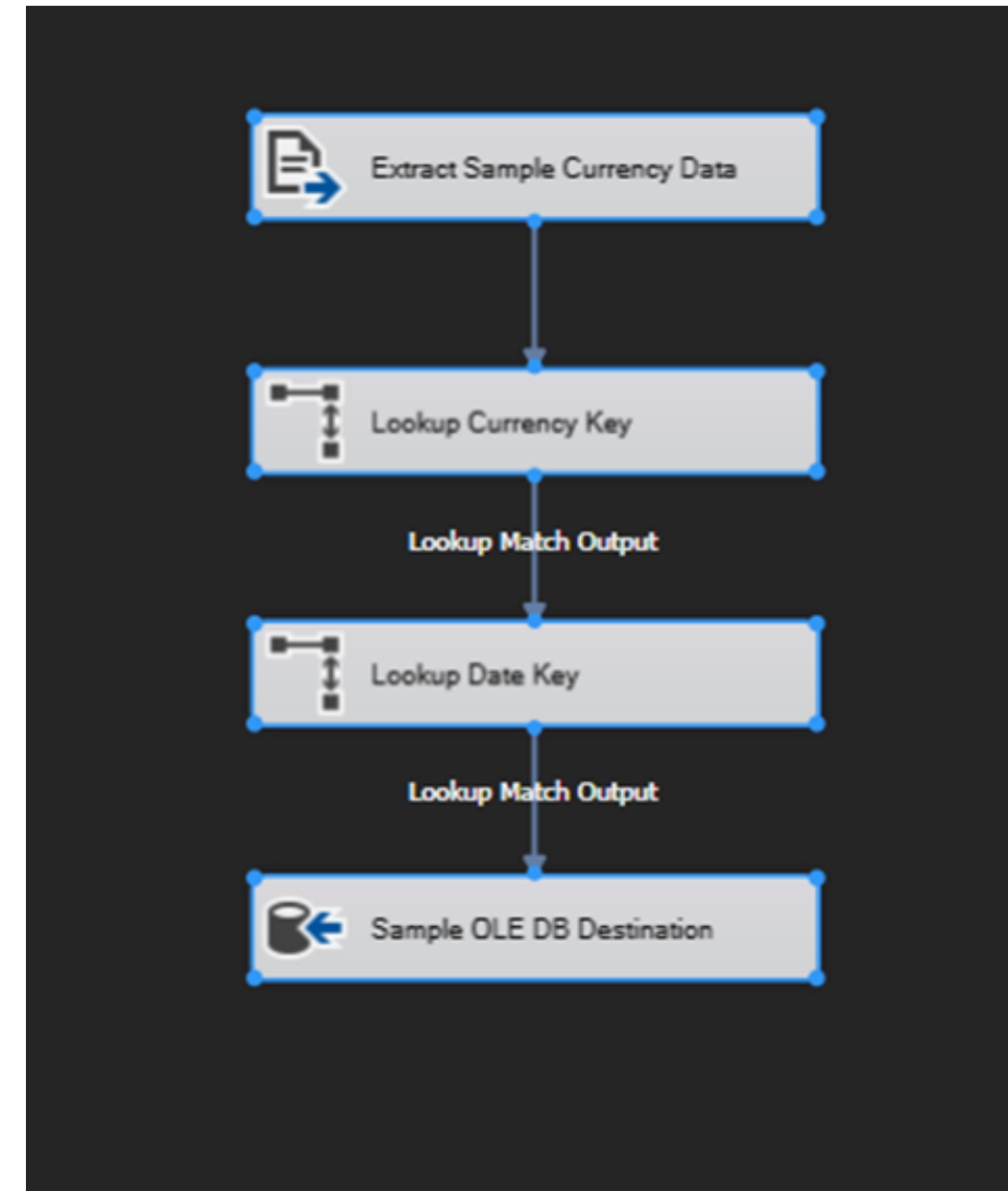
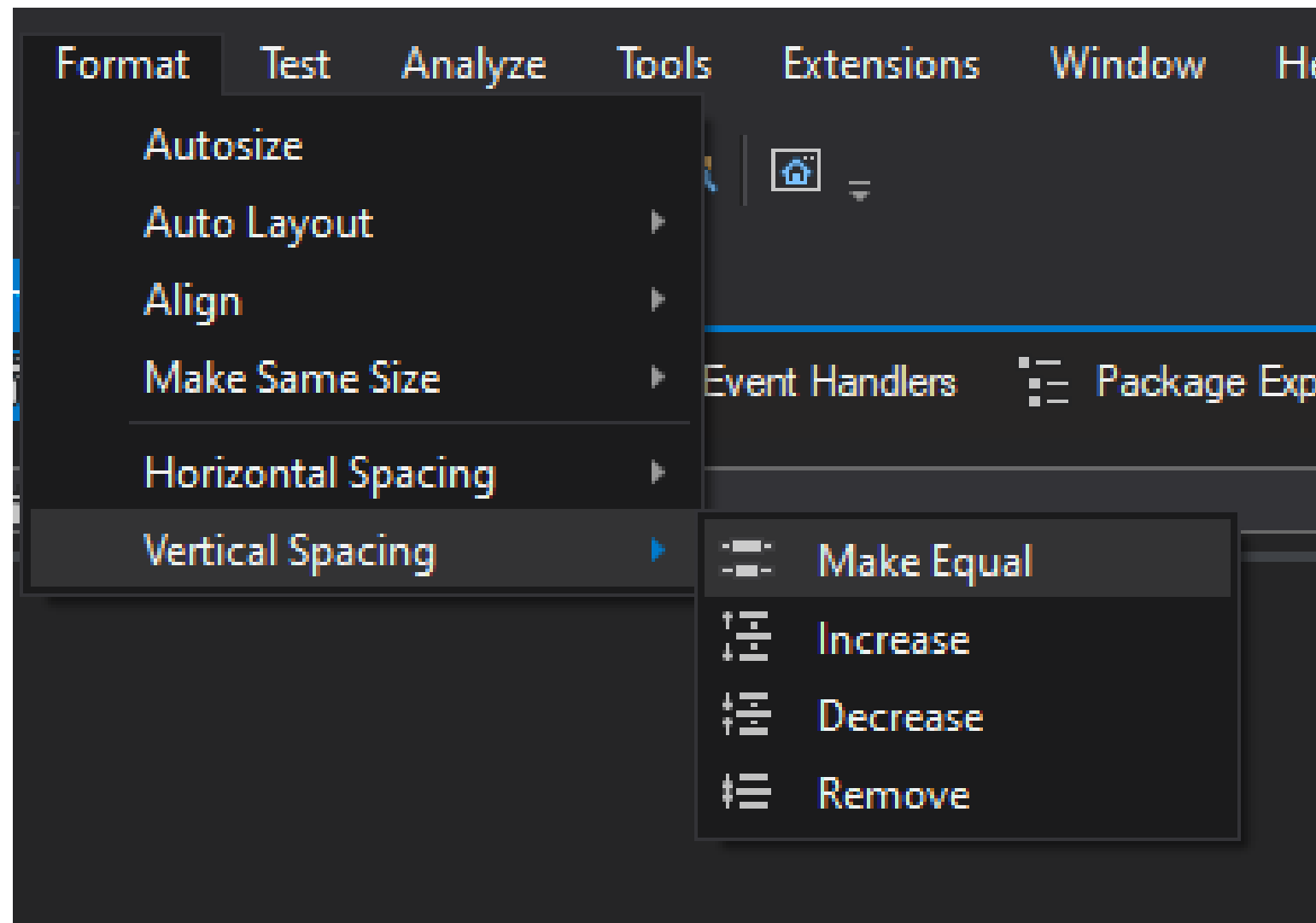
Lección 1-8



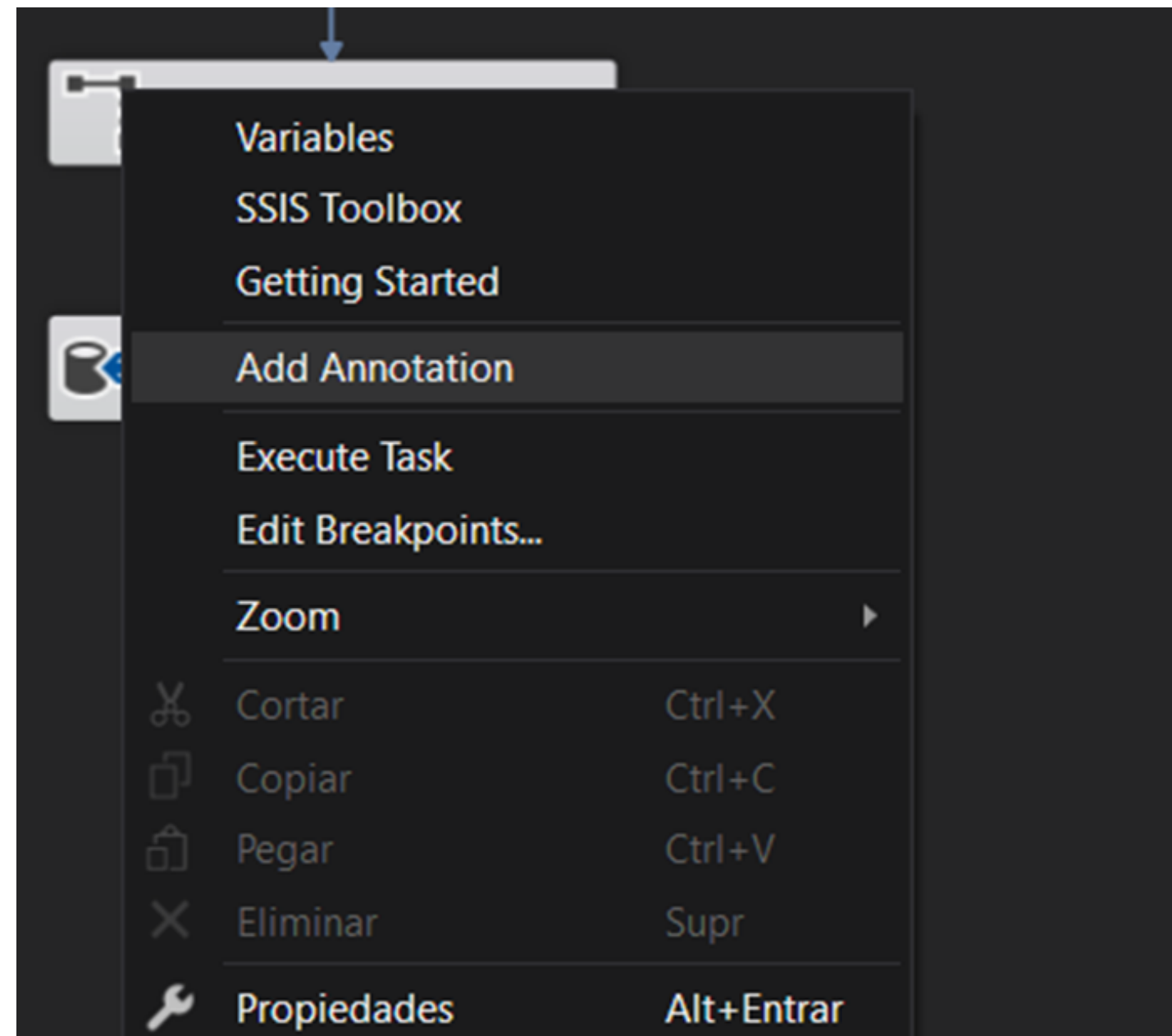
Lección 1-8



Lección 1-8



Lección 1-8



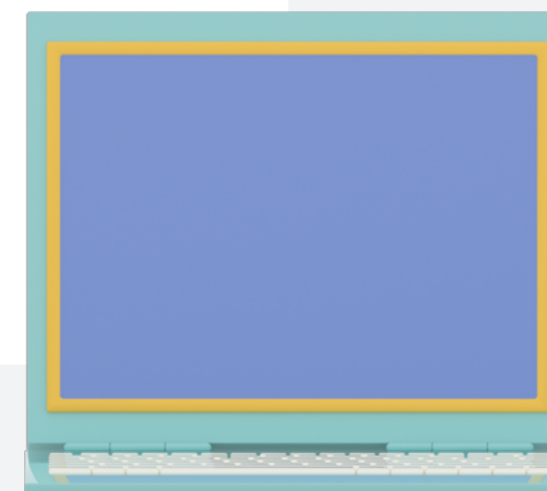
The data flow extracts data from a file, looks up values in the CurrencyKey column in the DimCurrency table and the DateKey column in the DimDate table, and writes the data to the NewFactCurrencyRate table.

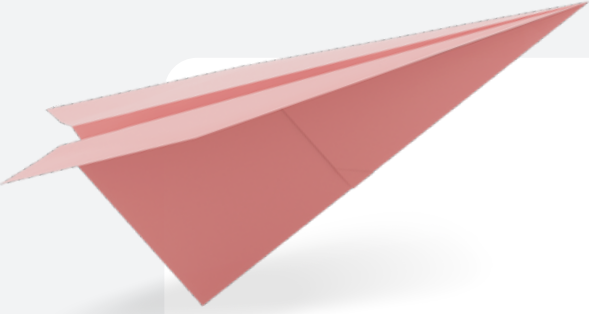
The data flow extracts data from a file, looks up values in the CurrencyKey column in the DimCurrency table and the DateKey column in the DimDate table, and writes the data to the NewFactCurrencyRate table.



A continuación...

Una demo





¿Tienes alguna pregunta?

Esperamos que hayas aprendido algo nuevo.

